Geometric Class Field Theory

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Abstract

We calculate the ramification of the symmetric product $\mathcal{F}^{[\deg \mathfrak{m}]}$ of a local system $\mathcal{F}$ on a curve $C \setminus \mathfrak{m}$. Assuming the ramification of $\mathcal{F}$ is bounded by $\mathfrak{m}$, we find that the symmetric product $\mathcal{F}^{[\deg \mathfrak{m}]}$ is at most tamely ramified at the generic point of the exceptional divisor $E_{\mathfrak{m}}$ of the blowup of $C^{(\deg \mathfrak{m})}$ at $\mathfrak{m}$. As a primary application, we utilize this result to prove Geometric Class Field Theory. Our approach builds upon the geometric framework for the unramified case originally established by Deligne for the rank-one Langlands correspondence, following the subsequent extensions to the ramified case developed by Takeuchi and Guignard.

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1 Introduction

In this thesis, we give an elementary proof of a certain imporant geometric theorem occuring in Deligne's approach to geometric class field theory. We (usually) work over a perfect field k, C is a projective smooth geometrically connected curve over k, with genus g. One of the main geometric ingredients in the approach, is showing why a local system $\mathcal{F}$ with ramification bounded by a modulus $\mathfrak{m} = \sum n_i P_i$ on $U = C \setminus \mathfrak{m}$ descends via the Abel-Jacobi $\Phi : U \to \mathrm{Pic}_{C,\mathfrak{m}}$ to $\mathrm{Pic}_{C,\mathfrak{m}}$. The approach, innovated by Deligne, relies on analyzing the symmetric powers $\mathcal{F}^{[d]}$ of $\mathcal{F}$ on the symmetric powers $U^{(d)}$ of U, and showing that for sufficiently large d, $\mathcal{F}^{[d]}$ descends to $\mathrm{Pic}^d_{C,\mathfrak{m}}$ via the degree d Abel-Jacobi map $\Phi_d : U^{(d)} \to \mathrm{Pic}^d_{C,\mathfrak{m}}$. The geometric-fibers of Φ_d (for $d \geq \deg \mathfrak{m} + 2g - 1$) over any point are isomorphic to

$$\begin{cases} \mathbb{A}^{d-\deg \mathfrak{m} - g + 1}_{k^{sep}} & \text{if } \mathfrak{m} > 0 \\ \mathbb{P}^{d-g}_{k^{sep}} & \text{if } \mathfrak{m} = 0 \end{cases}$$

Where g is the genus of the curve C. The unramified case ($\mathfrak{m} = 0$) is relatively simple, as the Abel-Jacobi map is proper, surjective with geometrically connected fibers, which follows from the fact that it is a fibration in projective spaces. Thus, by using the homotopy exact sequence for the étale fundamental group, one gets an isomorphism between the étale fundamental group of $U^{(d)} (= C^{(d)})$ and that of $\mathrm{Pic}^d_{C,\mathfrak{m}} (= \mathrm{Pic}^d_C)$.

The ramified case ($\mathfrak{m} > 0$) is more subtle, as the Abel-Jacobi map is not proper anymore, and one needs to analyze the ramification of $\mathcal{F}^{[d]}$ "along the boundary" of $U^{(d)}$ in $C^{(d)}$.

Previous work has generalized Deligne's approach to the ramified case, most notably by Guignard [Gui19] and Takeuchi [Tak19]. Their approaches differ. To descend, Guignard proves that the restriction of $\mathcal{F}^{[d]}$ to any line in the fiber of the degree d Abel-Jacobi map is a constant étale sheaf. He achieves this by demonstrating that the restriction is at most tamely ramified and invoking the triviality of the tame fundamental group of $\mathbb{A}^1_k$. His analysis relies on local geometric class field theory. It is also worth noting that Guignard's method generalizes to relative curves over arbitrary base schemes. Takeuchi, on the other hand, constructs a compactification of $U^{(d)}$ by blowing up $C^{(d)}$ along certain well-chosen centers. This compactification, denoted by $\tilde{C}^{(d)}_{\mathfrak{m}}$, has $U^{(d)}$ as an open subscheme with a codimension 1 closed subscheme H as complement. He then shows that the Abel-Jacobi map extends to a proper morphism from $\tilde{C}^{(d)}_{\mathfrak{m}}$ to $\mathrm{Pic}^d_{C,\mathfrak{m}}$, which is a fibration in projective spaces. Thus, by the homotopy exact sequence for the étale fundamental group, one gets an isomorphism between the étale fundamental group of $\tilde{C}^{(d)}_{\mathfrak{m}}$ and that of $\mathrm{Pic}^d_{C,\mathfrak{m}}$. To conclude the descent, Takeuchi analyzes the ramification of $\mathcal{F}^{[d]}$ along the boundary H of $\tilde{C}^{(d)}_{\mathfrak{m}}$, showing that it is tamely ramified there, which suffices. His methods relies on the theory of Witt vectors and refined Swan conductors.

For an account of these approaches, see [Gui19] and [Tak19]. For a full approach following Deligne's method in the unramified case, and the tamely ramified case see [Ten15], and [Tót11].

In this thesis, we calculate the ramification of $\mathcal{F}^{[d]}$ directly to show that it is at most tame at the generic point of H, avoiding the use of Swan conductors. To prove Geometric Class Field Theory, we utilize Takeuchi's construction of the compactification $\tilde{C}^{(d)}_{\mathfrak{m}}$ of $U^{(d)}$ via the blowing up of $C^{(d)}$.

In the rest of the introduction, we state the main theorem of geometric class field theory Theorem 1, and its reductions to Theorem 2 and Theorem 3, which we prove in this thesis.

Let k be a perfect field, and let C be a projective smooth geometrically connected curve over k, with genus g. Geometric class field theory gives a geometric description of abelian coverings of C by relating it to isogenies of the generalized Picard schemes.

Fix a modulus $\mathfrak{m}$, i.e. an effective Cartier divisor of C and let U be its complement in C. The pairs $(\mathcal{L}, \alpha)$, where $\mathcal{L}$ is an invertible $\mathcal{O}_C$-module and α is a rigidification of $\mathcal{L}$ along $\mathfrak{m}$, are parametrized by a k-group scheme $\mathrm{Pic}_{C,\mathfrak{m}}$, called the rigidified Picard scheme. The Abel-Jacobi morphism

$$\Phi : U \to \mathrm{Pic}_{C,\mathfrak{m}}$$

is the morphism which sends a section x of U to the pair $(\mathcal{O}(x), 1)$. The fundamental result of GCFT can be formulated as:

Theorem 1 (Geometric Class Field Theory)**.** *Let Λ be a finite ring of cardinality invertible in k, and let $\mathcal{F}$ be an étale sheaf of Λ-modules, locally free of rank 1 on U, with ramification bounded by $\mathfrak{m}$. Then, there exists a unique (up to isomorphism) multiplicative*[1] *étale sheaf of Λ-modules $\mathcal{G}$ on $\mathrm{Pic}_{C,\mathfrak{m}}$, locally free of rank 1, such that the pullback of $\mathcal{G}$ by Φ is isomorphic to $\mathcal{F}$.*

Let d be a positive integer. We denote by $U^{(d)}$ the d-th symmetric power of U over k. For an étale sheaf $\mathcal{F}$ on $U_{\text{ét}}$, we denote by $\mathcal{F}^{[d]}$ the d-th symmetric power of $\mathcal{F}$ on $U^{(d)}$ The degree d Abel-Jacobi morphism is defined as the map

$$\Phi_d : U^{(d)} \to \mathrm{Pic}^d_{C,\mathfrak{m}}$$

which sends a section $x_1 + \cdots + x_d$ of $U^{(d)}$ to the pair $(\mathcal{O}(x_1 + \cdots + x_d), 1)$.

A method of descent shows that to prove Theorem 1, it suffices to prove the following reduced version[2]:

Theorem 2. *Let Λ be a finite ring of cardinality invertible in k, and let $\mathcal{F}$ be an étale sheaf of Λ-modules, locally free of rank 1 on $U_{\acute{e}t}$, with ramification bounded by $\mathfrak{m}$. Then, for sufficiently large integer d, there exists a unique (up to isomorphism) étale sheaf of Λ-modules $\mathcal{G}_d$ on $\mathrm{Pic}^d_{C,\mathfrak{m}}$, locally free of rank 1, such that the pullback of $\mathcal{G}_d$ by Φ_d is isomorphic to $\mathcal{F}^{[d]}$.*

To prove Theorem 2 we follow the work of [Tak19] (and similiary done in [Tót11]), analyzing the ramification of $\mathcal{F}^{[d]}$ after blowing up $C^{(d)}$. We analyze this ramification using elementary methods. We prove the following Theorem:

Theorem 3. *Let Λ be a finite ring of cardinality invertible in k, and let $\mathcal{F}$ be an étale sheaf of Λ-modules, locally free of rank 1 on $U_{\acute{e}t}$, with ramification bounded by $\mathfrak{m} = \sum n_i P_i$. Let $\mathfrak{n} = n_i P_i \subset \mathfrak{m}$ be a sub modulus of $\mathfrak{m}$. Then $\mathcal{F}^{[\deg \mathfrak{n}]}$ is tamely ramified at the generic point $\eta_\mathfrak{n}$ of the exceptional divisor $E_\mathfrak{n}$ of the blowup $X_\mathfrak{n}$ of $C^{(\deg \mathfrak{n})}$ at the closed point $\mathfrak{n}$.*

The thesis is organized as follows:

Chapter 1 - Preliminaries provides the necessary preliminaries and covers the foundational material upon which this work is based, generally without providing proofs.

Chapter 2 - Ramification After Blowup Is Tame is devoted to the proof of Theorem 3, along with several corollaries that will be instrumental in the proof of Theorem 2.

[1]The notion of a multiplicative locally free Λ-module of rank 1 is due to [Gui19] and corresponds to isogenies $G \to \mathrm{Pic}_{C,\mathfrak{m}}$ with constant kernel $\Lambda^\times$. This concept corresponds to multiplicative characters of $H^1(\mathrm{Pic}_{C,\mathfrak{m}}, \mathbb{Q}/\mathbb{Z})$ in the formulation of [Tak19], and generalizes Hecke eigensheaves in the context of [Ten15].

[2]See the last page of [Gui19], Section 8.3 of [Ten15], or the proof of Theorem 1.2 in [Tak19] for details on this reduction.

Chapter 3 - Geometric Class Field Theory presents the proof of Theorem 2.

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2 Preliminaries

This section establishes the foundational definitions and theorems necessary for the remainder of this work. We focus specifically on the theory of G-torsors, which are central to our study due to their correspondence with locally free sheaves of rank 1. Subsequently, we review the relevant background on ramification theory from the existing literature. Finally, we conclude with several algebraic geometric remarks and notational conventions that will be employed implicitly throughout this thesis.

2.1 Torsors

In what follows, we largely adhere to the treatment of torsors and group objects found in published notes of Alex Youcis [You20]. Let $\mathscr{C} = (\mathcal{C}, J)$ be a site and let $\mathcal{E} = Sh(\mathscr{C})$ be the associated topos. Let $\mathcal{G}$ be a group object in $\mathcal{E}$. We denote by $\mathcal{G}\mathcal{E}$ the category of objects in $\mathcal{E}$ endowed with a left $\mathcal{G}$-action.

Definition 4. A **$\mathcal{G}$-torsor** in $\mathcal{E}$ is an object $\mathcal{P}$ of $\mathcal{G}\mathcal{E}$ satisfying the following conditions:

1. The structural morphism $\mathcal{P} \to 1$ is an epimorphism in $\mathcal{E}$ (i.e., $\mathcal{P}$ is locally non-empty).
2. The map $\mathcal{G} \times \mathcal{P} \to \mathcal{P} \times \mathcal{P}$ defined by $(g, p) \mapsto (g \cdot p, p)$ is an isomorphism in $\mathcal{E}$ (i.e., $\mathcal{G}$ acts simply transitively on $\mathcal{P}$).

Since $\mathcal{E}$ is the topos of sheaves on a site $\mathscr{C}$, the definition can be reformulated in terms of covers. A $\mathcal{G}$-sheaf $\mathcal{P}$ is a $\mathcal{G}$-torsor if:

1. For every object $X \in \mathcal{C}$, there exists a covering $\{U_i \to X\} \in J$ such that $\mathcal{P}(U_i) \neq \emptyset$ for all i.
2. For any $X \in \mathcal{C}$ where $\mathcal{P}(X)$ is non-empty, the action of $\mathcal{G}(X)$ on $\mathcal{P}(X)$ is simply transitively.

A fundamental property of torsors is their local triviality: a $\mathcal{G}$-sheaf $\mathcal{P}$ is a $\mathcal{G}$-torsor if and only if it is locally isomorphic to the trivial torsor. Specifically, for every $X \in \mathcal{C}$, there must exist a cover $\{U_i \to X\}$ such that the restriction $\mathcal{P}|_{U_i}$ is isomorphic, as a $\mathcal{G}|_{U_i}$-sheaf, to $\mathcal{G}|_{U_i}$ acting on itself by left multiplication.

A **morphism of $\mathcal{G}$-torsors** $f : \mathcal{P}_1 \to \mathcal{P}_2$ is a morphism of sheaves that is equivariant with respect to the $\mathcal{G}$-action.

It is a standard result that every morphism of $\mathcal{G}$-torsors is an isomorphism. Consequently, the category of $\mathcal{G}$-torsors in $\mathcal{E}$ forms a groupoid.

Definition 5. We denote the groupoid of $\mathcal{G}$-torsors in $\mathcal{E}$ by $\mathbf{Tors}(\mathcal{E},\mathcal{G})$. The set of isomorphism classes of $\mathcal{G}$-torsors is denoted by $\mathrm{Tors}(\mathcal{E},\mathcal{G})$.

Torsors exhibit functoriality with respect to the group object:

Definition 6. Let $\varphi : \mathcal{G}_1 \to \mathcal{G}_2$ be a morphism of group sheaves on $\mathscr{C}$, and let $\mathcal{P}$ be a $\mathcal{G}_1$-torsor. We define the **contracted product** $\mathcal{G}_2 \times^{\mathcal{G}_1} \mathcal{P}$ as the quotient sheaf $\mathcal{G}_1 \backslash (\mathcal{G}_2 \times \mathcal{P})$, where $\mathcal{G}_1$ acts on the product by:

$$g_1 \cdot (g_2, p) = (g_2\varphi(g_1)^{-1}, g_1 \cdot p)$$

The contracted product inherits a natural left $\mathcal{G}_2$-action given on local sections by $h \cdot [g_2, p] = [hg_2, p]$, which endows it with the structure of a $\mathcal{G}_2$-torsor.

This construction yields a functor:

$$\varphi_* : \mathbf{Tors}(\mathcal{E},\mathcal{G}_1) \to \mathbf{Tors}(\mathcal{E},\mathcal{G}_2), \quad \mathcal{P} \mapsto \mathcal{G}_2 \times^{\mathcal{G}_1} \mathcal{P}$$

On the level of isomorphism classes, φ_* induces a map of pointed sets $\mathrm{Tors}(\mathcal{E},\mathcal{G}_1) \to \mathrm{Tors}(\mathcal{E},\mathcal{G}_2)$, sending the class of the trivial $\mathcal{G}_1$-torsor to the class of the trivial $\mathcal{G}_2$-torsor.

When $\mathcal{G}$ is a **sheaf of abelian groups** (an *abelian sheaf*), the pointed set $\mathrm{Tors}(\mathcal{E},\mathcal{G})$ inherits the structure of an abelian group:

Definition 7. Assume $\mathcal{G}$ is abelian sheaf. Let $\mathcal{P}_1$ and $\mathcal{P}_2$ be objects of $\mathbf{Tors}(\mathcal{E},\mathcal{G})$. We define the **product** $\mathcal{P}_1 \otimes^{\mathcal{G}} \mathcal{P}_2$ to be the quotient sheaf $(\mathcal{P}_1 \times \mathcal{P}_2)/\mathcal{G}$, where $\mathcal{G}$ acts on T-points by having:

$$g \cdot (f_1, f_2) := (gf_1, g^{-1}f_2)$$

The group object $\mathcal{G}$ then acts on the resulting quotient via its action on the presheaf quotient, which is given on classes by:

$$g \cdot [(f_1, f_2)] = [(gf_1, f_2)] = [(f_1, gf_2)]$$

where the square brackets denote the class in the quotient set.[3]

Let $[\mathcal{P}_1]$ and $[\mathcal{P}_2]$ be the isomorphism classes of $\mathcal{P}_1$ and $\mathcal{P}_2$ in $\mathrm{Tors}(\mathcal{E},\mathcal{G})$. We define the sum $[\mathcal{P}_1] + [\mathcal{P}_2]$ to be the class $[\mathcal{P}_1 \otimes^{\mathcal{G}} \mathcal{P}_2]$. This structure turns $\mathrm{Tors}(\mathcal{E},\mathcal{G})$ into an abelian group, where the identity is the class of the trivial torsor and the inverse is obtained by the opposite action.

Let Λ be a commutative ring object in the topos $\mathcal{E}$. We denote by $\mathbf{Pic}(\mathcal{E},\Lambda)$ the groupoid of locally free Λ-modules of rank 1 in $\mathcal{E}$ (i.e., invertible modules and their isomorphisms). For the multiplicative group object $G = \Lambda^\times$, there exists a classical correspondence between such Λ-modules and G-torsors. Indeed, both categories are symmetric monoidal categories, $\mathbf{Pic}(\mathcal{E},\Lambda)$ under the tensor product $\otimes_\Lambda$, with unit object Λ. And $\mathbf{Tors}(\mathcal{E},G)$ under the product $\otimes^G$, with unit object G acting on itself by translation.

Throughout this thesis, we shall primarily adopt the language of G-torsors, with the understanding that the associated Λ-module structure is implicitly inferred.

The following proposition formalizes this dictionary:

Proposition 8. *The associated module functor $\Phi : \mathcal{P} \mapsto \mathcal{P} \otimes^G \Lambda$ induces a canonical equivalence of monoidal categories:*

$$\Phi : (\mathbf{Tors}(\mathcal{E},G), \otimes^G, G) \xrightarrow{\sim} (\mathbf{Pic}(\mathcal{E},\Lambda), \otimes_\Lambda, \Lambda)$$

[3]Equivantly, the sum is obtained as the contracted product of the $\mathcal{G} \times \mathcal{G}$-torsor $\mathcal{P}_1 \times \mathcal{P}_2$ along the multiplication map $m : \mathcal{G} \times \mathcal{G} \to \mathcal{G}$.

with quasi-inverse functor given by $\Psi : \mathcal{L} \mapsto \underline{\mathrm{Isom}}_\Lambda(\Lambda, \mathcal{L})$.

Lastly, for any object $\mathcal{X} \in \mathcal{E}$, there is a canonical identification between the slice category $(\mathcal{G}\mathcal{E})/\mathcal{X}$ and the category of group objects $\mathcal{G}(\mathcal{E}/\mathcal{X})$, where $\mathcal{X}$ is viewed as having the trivial $\mathcal{G}$-action.

We denote by $\mathbf{Tors}(\mathcal{X}, \mathcal{G})$ the category of G-torsors over $\mathcal{X}$ in $\mathcal{G}\mathcal{E}_{/\mathcal{X}}$.

2.2 Torsors over the Étale's Sites

Fix a scheme X, and let G be a smooth affine X-group scheme. Denote by $X_{\text{Ét}}$ and $X_{\text{ét}}$ the big and small étale sites on X, respectively. The functor of points of G, which we denote by $G(\cdot) = \mathrm{Hom}_X(-, G)$, defines a group sheaf on $X_{\text{ét}}$.

Definition 9. A *principal G-bundle* (or *principal homogeneous space*) for G is a scheme $f : Y \to X$ equipped with a left G-action $\rho : G \times_X Y \to Y$ satisfying:

1. The morphism $G \times_X Y \to Y \times_X Y$ sending $(g, y) \mapsto (gy, y)$ is an isomorphism of X-schemes.
2. There exists an étale covering $\{U_i \to X\}$ such that $Y_{U_i} \cong G_{U_i}$ as G-schemes, where G acts on itself by left translation.

Remark. Since $G \to X$ is smooth and Y is étale-locally isomorphic to G, the morphism $f : Y \to X$ is smooth and affine. Note that Y is generally not étale over X unless G is an étale group scheme (e.g., a finite constant group). Similarly, if G is finite then $Y \to X$ is finite.

Similarly to the case of G-torsors of a topos, we define a morphism of principal G-bundles to be a morphism of X-schemes commuting with the G-action. We now have:

Theorem 10. *The morphism sending $Y \mapsto \mathrm{Hom}_X(-, Y)$ is an equivalence of categories from the category of principal G-bundles to the category of G-torsors on $X_{\acute{E}t}$ Similarly, the morphism sending Y to $\mathrm{Hom}_X(-, Y)$ to the category of G-torsors on $X_{\acute{e}t}$ is an equivalence.*

Corollary 11. *There is a natural equivalence $\mathbf{Tors}(X_{\acute{e}t}, G) \cong \mathbf{Tors}(X_{\acute{E}t}, G)$ inducing a bijection of pointed sets $\mathrm{Tors}(X_{\acute{e}t}, G) \xrightarrow{\cong} \mathrm{Tors}(X_{\acute{E}t}, G)$ which is an isomorphism of abelian groups if G is abelian. And every G-torsor, in each of the above sites can be realized as a scheme, which is a prinicipal G-bundle.*

Constant Finite Group Torsors

Let G be a finite group, within this section, we denote the associated constant group scheme over X by $\underline{G}$ to maintain a clear distinction between the group as a set and the group as a scheme. In subsequent sections, we shall follow the standard practice of identifying G with its constant group scheme and omit the underline for brevity.

$\underline{G}$ is given by $\coprod_{g \in G} X$ with the action shuffeling the X's according to multiplication.

By following the definitions, one sees that if X be a connected scheme and $f : Y \to X$ is a finite Galois cover with Galois group G. Then, $f : Y \to X$ is a principal $\underline{G}$-bundle. (Recall that a *finite Galois cover* is a finite étale surjection $Y \to X$ with Y connected and such that $G = \mathrm{Aut}(Y/X)$ acts transitively on the geometric points of Y lying over any geometric point of X).

On the otherhand if $f : Y \to X$ is a principal $\underline{G}$-bundle with Y connected, then Y is a finite Galois cover with automorphism group G.

However, not all $\underline{G}$-torsors are connected. If $H \subset G$ is a proper subgroup then any connected finite étale cover $f : Y \to X$ with Galois group H gives rise to a non-connected $\underline{G}$-torsor by looking at the induced $\underline{G}$-torsor $\varphi_*(Y)$ under the inclusion $\varphi : \underline{H} \to \underline{G}$. On the otherhand, if we fix a geometric point $\overline{x} \to X$, then to give an homomorphism $\rho \in \mathrm{Hom}_{\mathrm{cont}}(\pi_1^{\mathrm{et}}(X, x), G)$ is equivilant to give a connected pointed Galois cover $(Y, \overline{y}) \to (X, \overline{x})$ with Galois group $H = \rho(\pi_1^{\mathrm{et}}(X, x)) \subset G$. Thus, pushing forward to $\underline{G}$ we get a prinicipal $\underline{G}$-bundle. The choice of a different geometric point $\overline{x'} \to X$ differ the homomorphism by an inner automorphism, thus we have:

Theorem 12. *Let X be a connected scheme and $\overline{x}$ a geometric point of X. Suppose in addition that G is a finite abstract group. Define a map*

$$\mathrm{Hom}_{\mathrm{cont}}(\pi_1^{\mathrm{et}}(X, \overline{x}), G)/\mathrm{Inn}(G) \to \mathrm{Tors}(X_{\acute{E}t}, \underline{G}) \tag{1}$$

by sending a homomorphism $\rho : \pi_1^{\mathrm{et}}(X, \overline{x}) \to G$ to the principal $\underline{G}$-bundle $\varphi_*(Y)$ where Y is the principal $\underline{\rho(\pi_1^{\mathrm{et}}(X, \overline{x}))}$-bundle obtained above and φ is the inclusion $\rho(\pi_1^{\mathrm{et}}(X, \overline{x})) \hookrightarrow G$. Then, the map is a bijection of pointed sets where the trivial homomorphisms (which is the only element of its $\mathrm{Inn}(G)$*-orbit) is the distinguished element of the left hand side.*

If G is abelian then $\mathrm{Inn}(G)$ is trivial, and we obtain:

Corollary 13. *Let G be a finite abelian group, X a connected scheme, and $\overline{x}$ a geometric point of X. Then, the map from Theorem 12 induces an isomorphism of abelian groups*

$$\mathrm{Hom}_{\mathrm{cont.}}(\pi_1^{\acute{e}t}(X, \overline{x}), G) \xrightarrow{\cong} \mathrm{Tors}(X_{\acute{E}t}, \underline{G}) \tag{2}$$

Decomposition of Torsors

In the case of finite groups, the functoriality of the contracted product has a concrete geometric interpretation as a tower of covers. Let G be a finite abelian group and $H \subseteq G$ a subgroup. Let $\pi : G \to G/H$ be the natural quotient map.

Proposition 14. *Let $\mathcal{P} \to X$ be a G-torsor in $X_{\acute{e}t}$. There exists a natural factorization of the morphism $\mathcal{P} \to X$:*

$$\mathcal{P} \xrightarrow{\phi} \mathcal{P}_{G/H} \xrightarrow{\psi} X$$

where:

1. *$\mathcal{P}_{G/H} := \mathcal{P} \times^G (G/H)$ is a G/H-torsor over X.*
2. *$\mathcal{P}$ is an H-torsor over the scheme $\mathcal{P}_{G/H}$.*

Proof. The first assertion follows directly from the definition of the contracted product functor $\pi_* :$ $\mathbf{Tors}(X, G) \to \mathbf{Tors}(X, G/H)$. To see the second assertion, we note that the map $\phi : \mathcal{P} \to \mathcal{P}_{G/H}$ is surjective. Locally on X, we may choose a cover $U \to X$ such that $\mathcal{P}|_U \cong G \times U$. Over this cover, the map ϕ is identified with the product of the quotient map and the identity:

$$\pi \times \mathrm{id}_U : G \times U \to (G/H) \times U$$

Since $G \to G/H$ is a trivial H-torsor (the fiber over any element $\bar{g} \in G/H$ is a coset gH, which is an H-set isomorphic to H acting on itself), $\mathcal{P}$ is locally an H-torsor over $\mathcal{P}_{G/H}$. By the descent of torsors, $\mathcal{P}$ is an H-torsor over $\mathcal{P}_{G/H}$ globally. $\square$

2.3 Symmetric Powers of Schemes and Torsors

This section reviews the construction of quotients for schemes and torsors under finite group actions, specifically focusing on symmetric powers. To ensure these quotients exist as schemes, we utilize the framework of admissible actions from [SGA1]. Our treatment here closely follows the exposition in [Gui19] The definitions and results presented below are adapted from their work, the difference being that we ask Γ to act on the left, to make the later construction of symmetric powers more natural. This foundation provides the necessary criteria for admissibility and base change required to define the symmetric powers of a scheme X and a G-torsor $\mathcal{P}$ over X.

Let S be a scheme.

Definition 15 (([SGA1], V.1.7).)**.**

- Let T be an object of a category $\mathcal{C}$ endowed with a right action of a group Γ. We say that **the quotient** T/Γ **exists** in $\mathcal{C}$ if the covariant functor

$$\begin{aligned} \mathcal{C} &\to \mathrm{Sets} \\ U &\mapsto \mathrm{Hom}_{\mathcal{C}}(T, U)^{\Gamma} \end{aligned}$$

  is representable by an object of $\mathcal{C}$.

- Let T be an S-scheme. An action of a finite group Γ on T is **admissible** if there exists an affine Γ-invariant morphism $f : T \to T'$ such that the canonical morphism $\mathcal{O}_{T'} \to f_*\mathcal{O}_T$ induces an isomorphism from $\mathcal{O}_{T'}$ to $(f_*\mathcal{O}_T)^{\Gamma}$.

Proposition 16. *The following holds:*

1. ***([SGA1] V.1.3).*** *Let T be an S-scheme endowed with an admissible right action of a finite group Γ. If $f : T \to T'$ is an affine Γ-invariant morphism such that the canonical morphism $\mathcal{O}_{T'} \to f_*\mathcal{O}_T$ induces an isomorphism from $\mathcal{O}_{T'}$ to $(f_*\mathcal{O}_T)^{\Gamma}$, then the quotient T/Γ exists and is isomorphic to T'.*

2. ***([SGA1], V.1.8).*** *Let T be an S-scheme endowed with a right action of a finite group Γ. Then, the action of Γ on T is admissible if and only if T is covered by Γ-invariant affine open subsets.*

3. ***([SGA1], V.1.9).*** *Let T be an S-scheme endowed with an admissible right action of a finite group Γ, and let S' be a flat S-scheme. Then, the action of Γ on the S'-scheme $T \times_S S'$ is admissible, and the canonical morphism*

$$(T \times_S S')/\Gamma \to (T/\Gamma) \times_S S'$$

   is an isomorphism.

Now let X be an S-scheme and let $d \geq 0$ be an integer. The group S_d of permutations of $[\![1, d]\!]$ acts on the right on the S-scheme $X^{\times_S d} = X \times_S \cdots \times_S X$ by the formula

$$(x_i)_{i \in [\![1,d]\!]} \cdot \sigma = (x_{\sigma(i)})_{i \in [\![1,d]\!]}.$$

Proposition 17 ([Gui19] Proposition 2.27)**.** *If X is a scheme, Zariski locally quasi-projective over S, then the right action of the symmetric group S_d on the d-fold fiber product $X^{\times_S d}$ is admissible. Consequently, the quotient $\mathrm{Sym}^d_S(X) = X^{\times_S d}/S_d$ exists as a scheme over S.*

Definition 18. Under the hypotheses of the proposition above, we define the **relative symmetric product** of X over S of degree d as the quotient

$$\mathrm{Sym}^d_S(X) \coloneqq X^{\times_S d}/S_d.$$

When the base scheme S is clear from the context, we shall denote this quotient by $X^{(d)}$.

Guingard shows that when $X = \mathrm{Spec}(B)$ and $S = \mathrm{Spec}(A)$ then $\mathrm{Sym}^d_S(X)$ is representable by an affine S-scheme (See [Gui19] Remark 2.28).

Proposition 19 ([Gui19] Proposition 2.28)**.** *If X is flat and Zariski-locally quasi-projective over S, then $Sym^d_S(X)$ is flat over S. Moreover, for any S-scheme S', the canonical morphism*

$$Sym^d_{S'}(X \times_S S') \to Sym^d_S(X) \times_S S'$$

is an isomorphism.

Symmetric Powers of Torsors

Let G be a finite abelian group, let $\mathcal{P}$ be a G-torsor over an S-scheme X in $S_{\text{Ét}}$.

Proposition 20 ([[SGA1]], IX.5.8)**.** *Assume that $\mathcal{P}$ and X are endowed with right actions from a finite group Γ such that the morphism $\mathcal{P} \to X$ is Γ-equivariant, and that the following properties hold:*

(a) The right Γ-action on $\mathcal{P}$ commutes with the left G-action.

(b) The right Γ-action on X is admissible, and the quotient morphism $X \to X/\Gamma$ is finite.

(c) For any geometric point $\bar{x}$ of X, the action of the stabilizer $\Gamma_{\bar{x}}$ of $\bar{x}$ in Γ on the fiber $\mathcal{P}_{\bar{x}}$ of $\mathcal{P}$ at $\bar{x}$ is trivial.

Then the action of Γ on $\mathcal{P}$ is admissible, and $\mathcal{P}/\Gamma$ is a G-torsor over X/Γ in $S_{\acute{E}t}$.

By Theorem 10, $\mathcal{P}$ is representable by a finite étale X-scheme. For each $i \in [\![1, d]\!]$ let $p_i : X^{\times_S d} \to X$ be the projection on i-th factor, and let us consider the G-torsor

$$p_1^{-1}\mathcal{P} \otimes^G \cdots \otimes^G p_d^{-1}\mathcal{P} = K_G \backslash \mathcal{P}^{\times_S d}$$

over $X^{\times_S d}$, where $K_G \subseteq G^d$ is the kernel of the multiplication morphism $G^d \to G$ (the dependence on d is suppressed in the notation). [4] The object $K_G \backslash \mathcal{P}^{\times_S d}$ of $S_{\text{Ét}}$ is too representable by an S-scheme which is finite étale over $X^{\times_S d}$. The group S_d acts on the right on $K_G \backslash \mathcal{P}^{\times_S d}$ by the formula

$$(p_i)_{i \in [\![1,d]\!]} \cdot \sigma = (p_{\sigma(i)})_{i \in [\![1,d]\!]}.$$

This action of S_d commutes with the left action of G on $K_G \backslash \mathcal{P}^{\times_S d}$.

Proposition 21 ([Gui19] Proposition 2.32.)**.** *If X is Zariski-locally quasi-projective on S, then the right action of S_d on $K_G \backslash \mathcal{P}^{\times_S d}$ is admissible, so that the quotient $\mathcal{P}^{[d]}$ of $K_G \backslash \mathcal{P}^{\times_S d}$ by S_d exists as a scheme over S. Moreover, the canonical morphism $\mathcal{P}^{[d]} \to \mathrm{Sym}^d_S(X)$ is a G-torsor, and the morphism*

$$p_1^{-1}\mathcal{P} \otimes^G \cdots \otimes^G p_d^{-1}\mathcal{P} \to r^{-1}\mathcal{P}^{[d]}$$

where $r : X^{\times_S d} \to \mathrm{Sym}^d_S(X)$ is the canonical projection, is an isomorphism of G-torsors over $X^{\times_S d}$.

[4] Equivantly, the sum is obtained as the contracted product of the G^d-torsor $p_1^{-1}\mathcal{P}_1 \times \cdots \times p_d^{-1}\mathcal{P}_d$ along the multiplication map $m : G^d \to G$.

When G is abelian, and letting the S_d-action be a left action (via the inverse), then the condition $\prod_{i=1}^d g_i = 1$ defining K_G is invariant under permuting the coordinates. Therefore, the left action of S_d normalizes the action of K_G. The group generated by these two actions is the semidirect product $\Xi = K_G \rtimes S_d$, which is a finite group acting on the left on $\mathcal{P}^d$. Then the double qoutient $(K_G \backslash \mathcal{P}^{\times_S d})/S_d \cong S_d \backslash (K_G \backslash \mathcal{P}^{\times_S d})$ is the same as the quotient $\Xi \backslash \mathcal{P}^{\times_S d}$, where Ξ acts on $\mathcal{P}^{\times_S d}$ by the formula:

$$((g_1, \dots, g_d), \sigma) \star (p_1, \dots, p_d) = (g_1 p_{\sigma^{-1}(1)}, g_2 p_{\sigma^{-1}(2)}, \dots, g_d p_{\sigma^{-1}(d)})$$

Corollary 22. *Under the assumptions of the above discussion, there is a natural canonical morphism*

$$q : S_d \backslash \mathcal{P}^{\times_S d} \to \Xi \backslash \mathcal{P}^{\times_S d}$$

or equivlantly

$$q : S_d \backslash \mathcal{P}^{\times_S d} \to (K_G \backslash \mathcal{P}^{\times_S d})/S_d$$

between the symmetric power of $\mathcal{P}$ as a scheme and the symmetric power of $\mathcal{P}$ as a G-torsor. We make it a convention to denote the symmetric power of $\mathcal{P}$ as a scheme by $\mathcal{P}^{(d)}$, and the symmetric power of $\mathcal{P}$ as a G-torsor by $\mathcal{P}^{[d]}$. So that the above qoutient map is also written as: $q : \mathcal{P}^{(d)} \to \mathcal{P}^{[d]}$.

Proof. S_d is a subgroup of Ξ, hence any map that is Ξ-invariant is automatically S_d-invariant. $\square$

Remark. Note that the above discussion does not require that $\mathcal{P}$ is a G-torsor over X, only that it is a scheme with a left G-action.

Quotient of Torsors Towers

We now study how the symmetric power construction of Proposition 21 interacts with the canonical decomposition of Proposition 14. Throughout this subsection, G denotes a finite abelian group, $H \subseteq G$ a subgroup, and $\mathcal{P} \to X$ a G-torsor over an S-scheme X that is Zariski-locally quasi-projective over S. Recall that $\mathcal{P} \to X$ factors canonically as

$$\mathcal{P} \xrightarrow{\phi} \mathcal{P}_{G/H} \xrightarrow{\psi} X,$$

where $\mathcal{P}_{G/H} = \mathcal{P} \times^G (G/H)$ is a G/H-torsor over X and ϕ is an H-torsor.

In addition to the kernel $K_G = \ker(G^d \to G)$ of the summation morphism appearing in Proposition 21, we shall need the analogous *summation kernels*

$$K_H := \ker(H^d \to H), \qquad K_{G/H} := \ker((G/H)^d \to G/H).$$

Applying the snake lemma to the commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & H^d \arrow[r] \arrow[d, two heads] & G^d \arrow[r] \arrow[d, two heads] & (G/H)^d \arrow[r] \arrow[d, two heads] & 0 \\
0 \arrow[r] & H \arrow[r] & G \arrow[r] & G/H \arrow[r] & 0,
\end{tikzcd}$$

in which the vertical maps are the surjective summation morphisms, yields a short exact sequence

$$0 \to K_H \to K_G \to K_{G/H} \to 0. \tag{3}$$

The induced isomorphism $K_G/K_H \cong K_{G/H}$ will play a central role below.

We define four quotient schemes

$$
\begin{aligned}
\mathcal{P}^{[d]_G} &:= (K_G \rtimes S_d)\backslash \mathcal{P}^{\times_S d}, \\
\mathcal{P}^{[d]_H} &:= (K_H \rtimes S_d)\backslash \mathcal{P}^{\times_S d}, \\
(\mathcal{P}_{G/H})^{(d)} &:= (H^d \rtimes S_d)\backslash \mathcal{P}^{\times_S d}, \\
(\mathcal{P}_{G/H})^{[d]} &:= ((H^d + K_G) \rtimes S_d)\backslash \mathcal{P}^{\times_S d},
\end{aligned}
$$

where S_d acts on $\mathcal{P}^{\times_S d}$ as in Corollary 22 and the subgroups $K_H \subseteq K_G \subseteq H^d + K_G$ and $H^d \subseteq H^d + K_G$ of G^d act diagonally on $\mathcal{P}^{\times_S d}$ via the G-action on $\mathcal{P}$. The first space is the symmetric G-power of $\mathcal{P}$ as in Proposition 21; the remaining three are, respectively, the symmetric H-power of $\mathcal{P}$ relative to ϕ, the symmetric power of the scheme $\mathcal{P}_{G/H}$, and the symmetric G/H-power of $\mathcal{P}_{G/H}$. The latter two identifications follow from $(\mathcal{P}_{G/H})^{\times_S d} \cong H^d \backslash \mathcal{P}^{\times_S d}$ together with the fact that the preimage of $K_{G/H} \subseteq (G/H)^d$ under $G^d \twoheadrightarrow (G/H)^d$ equals $H^d + K_G$. the last isomorphism like that, is from the correspondence theorem, claude generated at the end of the article, add it somewhere

Proposition 23. *The four quotient schemes above exist and fit into a canonical commutative diagram*

$$
\begin{tikzcd}
\mathcal{P}^{[d]_H} \arrow[r, "H"] \arrow[d] & (\mathcal{P}_{G/H})^{(d)} \arrow[d, "q"] \\
\mathcal{P}^{[d]_G} \arrow[r, "H"'] \arrow[dr, bend right, "G"'] & (\mathcal{P}_{G/H})^{[d]} \arrow[d, "G/H"] \\
& X^{(d)},
\end{tikzcd}
$$

in which each labelled arrow is a torsor under the indicated constant group. The right-hand vertical q is the canonical projection of Corollary 22, applied to the G/H-torsor $\mathcal{P}_{G/H} \to X$; in general it is not *a torsor under any constant group scheme (see the remark below). The unlabelled left-hand vertical $\mathcal{P}^{[d]_H} \to \mathcal{P}^{[d]_G}$ is the canonical projection induced by the inclusion of S_d-stable subgroups $K_H \rtimes S_d \subseteq K_G \rtimes S_d$ in $G^d \rtimes S_d$; it is the base change of q along the structural map $\mathcal{P}^{[d]_G} \to (\mathcal{P}_{G/H})^{[d]}$ via the Cartesian property below, and likewise fails to be a torsor under a constant group scheme in general.*

Moreover, the upper square is Cartesian:

$$
\mathcal{P}^{[d]_H} \cong \mathcal{P}^{[d]_G} \times_{(\mathcal{P}_{G/H})^{[d]}} (\mathcal{P}_{G/H})^{(d)}.
$$

Proof. The four labelled arrows are torsors under the indicated constant groups by direct application of Proposition 21 and Proposition 14:

- Applied to the G-torsor $\mathcal{P} \to X$, Proposition 21 gives the G-torsor $\mathcal{P}^{[d]_G} \to X^{(d)}$ (the diagonal arrow).
- Applied to the H-torsor $\phi : \mathcal{P} \to \mathcal{P}_{G/H}$ (taking $\mathcal{P}_{G/H}$ in place of the base X), Proposition 21 gives the H-torsor $\mathcal{P}^{[d]_H} \to (\mathcal{P}_{G/H})^{(d)}$ (the top arrow).

- Applied to the G/H-torsor $\psi : \mathcal{P}_{G/H} \to X$, Proposition 21 gives the G/H-torsor $(\mathcal{P}_{G/H})^{[d]} \to X^{(d)}$ (the bottom-right vertical), and Corollary 22 produces the canonical morphism $q : (\mathcal{P}_{G/H})^{(d)} \to (\mathcal{P}_{G/H})^{[d]}$.

The factorization $\mathcal{P}^{[d]_G} \xrightarrow{H} (\mathcal{P}_{G/H})^{[d]} \xrightarrow{G/H} X^{(d)}$ of the diagonal arrow into an H-torsor followed by a G/H-torsor is the canonical decomposition of Proposition 14, applied to the G-torsor $\mathcal{P}^{[d]_G} \to X^{(d)}$ and the subgroup $H \subseteq G$. Indeed, both $\mathcal{P}^{[d]_G} \times^G (G/H)$ and $(\mathcal{P}_{G/H})^{[d]}$ are realized as the admissible quotient $((H^d + K_G) \rtimes S_d)\backslash \mathcal{P}^{\times_S d}$.

To prove the upper square is Cartesian, set $F := \mathcal{P}^{[d]_G} \times_{(\mathcal{P}_{G/H})^{[d]}} (\mathcal{P}_{G/H})^{(d)}$. Base-changing the H-torsor $\mathcal{P}^{[d]_G} \to (\mathcal{P}_{G/H})^{[d]}$ along q exhibits F as an H-torsor over $(\mathcal{P}_{G/H})^{(d)}$. The universal property of the fiber product furnishes a canonical morphism $\mathcal{P}^{[d]_H} \to F$ over $(\mathcal{P}_{G/H})^{(d)}$, which is H-equivariant since the H-actions on both source and target are induced by the diagonal H-action on $\mathcal{P}^{\times_S d}$. Any morphism of H-torsors is an isomorphism, so $\mathcal{P}^{[d]_H} \cong F$ and the upper square is Cartesian. The left vertical $\mathcal{P}^{[d]_H} \to \mathcal{P}^{[d]_G}$ is then identified, via this Cartesian square, with the base change of q along $\mathcal{P}^{[d]_G} \to (\mathcal{P}_{G/H})^{[d]}$. □

Remark. The two vertical morphisms $q : (\mathcal{P}_{G/H})^{(d)} \to (\mathcal{P}_{G/H})^{[d]}$ and $\mathcal{P}^{[d]_H} \to \mathcal{P}^{[d]_G}$ are not, in general, torsors under any constant group scheme. The obstruction is purely group-theoretic: in $K_G \rtimes S_d$ the subgroup $K_H \rtimes S_d$ is not normal, because the conjugation action of S_d on K_G by coordinate permutation is non-trivial — for $k \in K_G$ and $\sigma \in S_d$, the components of $k - \sigma \cdot k$ are $k_i - k_{\sigma^{-1}(i)} \in G$, which need not lie in H. Equivalently, $H^d \rtimes S_d$ is not normal in $(H^d + K_G) \rtimes S_d$. So the naive set-theoretic quotients $(K_G \rtimes S_d)/(K_H \rtimes S_d)$ and $((H^d + K_G) \rtimes S_d)/(H^d \rtimes S_d)$ are not groups, even though $K_G/K_H \cong K_{G/H}$ and $(H^d + K_G)/H^d \cong K_{G/H}$ as abstract groups.

A precise description of the structure these morphisms carry would require concepts not introduced in this thesis, namely:

- the *free locus* $\mathcal{P}^{[d],\circ} \subseteq \mathcal{P}^{[d]}$, defined as the preimage of the complement of the big diagonal $\Delta \subseteq X^{(d)}$ (the open locus on which the residual S_d-action is free);
- *twisted forms* of a constant finite group, obtained as the contracted product $T \times^{S_d} A$ of an étale S_d-torsor T with a finite group A carrying an S_d-action; this is a finite étale group scheme, étale-locally isomorphic to the constant group $\underline{A}$ but not globally so when the S_d-action on A is non-trivial and T does not split;
- *torsors under non-constant group schemes*, generalising the constant-group torsors used throughout this thesis.

With these in hand, one shows that, restricted to the free locus, q is a torsor under a twisted form of the constant group $K_{G/H}$, of degree $|K_{G/H}| = |G/H|^{d-1}$; the left vertical inherits the same description by base change. We do not need this refinement in what follows: only the four labelled torsors and the Cartesian property are used.

Proposition 23 encodes two distinct factorizations of $\mathcal{P}^{[d]_H} \to X^{(d)}$, which we shall refer to as the two *towers*:

The H-symmetric tower, obtained by traversing the top edge and the right column of the diagram:

$$\mathcal{P}^{[d]_H} \xrightarrow{H} (\mathcal{P}_{G/H})^{(d)} \xrightarrow{q} (\mathcal{P}_{G/H})^{[d]} \xrightarrow{G/H} X^{(d)};$$

The G-symmetric tower, obtained by traversing the left column and the bottom row of the diagram:

$$\mathcal{P}^{[d]_H} \longrightarrow \mathcal{P}^{[d]_G} \xrightarrow{H} (\mathcal{P}_{G/H})^{[d]} \xrightarrow{G/H} X^{(d)}.$$

In each tower the labelled arrows are torsors under the indicated constant groups; the unlabelled arrow and the morphism q are not, in general, torsors under a constant group scheme.

The Cartesian property in Proposition 23 expresses that the connection morphism $\mathcal{P}^{[d]_H} \to \mathcal{P}^{[d]_G}$ on the left is the base change of $q : (\mathcal{P}_{G/H})^{(d)} \to (\mathcal{P}_{G/H})^{[d]}$ along the structural map $\mathcal{P}^{[d]_G} \to (\mathcal{P}_{G/H})^{[d]}$. Properties of q that are stable under base change therefore descend automatically to the connection morphism on the left.

Compatibility with Addition

The construction of symmetric powers is compatible with a natural addition law. For any integers $d_1, d_2 \geq 0$, there exists a canonical **addition morphism**

$$+_{d_1,d_2} : X^{(d_1)} \times_S X^{(d_2)} \to X^{(d_1+d_2)}$$

induced by the equivariant isomorphism $X^{\times_S d_1} \times_S X^{\times_S d_2} \cong X^{\times_S (d_1+d_2)}$ with respect to the inclusion of the product of symmetric groups $S_{d_1} \times S_{d_2} \subseteq S_{d_1+d_2}$.

Proposition 24. *Let $\mathcal{P}$ be a G-torsor over X. There is a canonical isomorphism of G-torsors over $X^{(d_1)} \times_S X^{(d_2)}$:*

$$+_{d_1,d_2}^{-1}\left(\mathcal{P}^{[d_1+d_2]}\right) \cong r_1^{-1}\mathcal{P}^{[d_1]} \otimes^G r_2^{-1}\mathcal{P}^{[d_2]}.$$

Proof. Let $\pi = r_{d_1} \times r_{d_2}$. By the commutativity of the following diagram

$$\begin{tikzcd} X^{\times_S d_1} \times_S X^{\times_S d_2} \arrow[r, "\sim"] \arrow[d, "\pi"'] & X^{\times_S (d_1+d_2)} \arrow[d, "r_{d_1+d_2}"] \\ X^{(d_1)} \times_S X^{(d_2)} \arrow[r, "+_{d_1,d_2}"] & X^{(d_1+d_2)} \end{tikzcd}$$

and applying Proposition 21, we obtain canonical isomorphisms:

$$\pi^{-1}\left(+_{d_1,d_2}^{-1}\mathcal{P}^{[d_1+d_2]}\right) \cong r_{d_1+d_2}^{-1}\mathcal{P}^{[d_1+d_2]} \cong \bigotimes_{i=1}^{d_1+d_2}{}^{G} p_i^{-1}\mathcal{P}$$

and

$$\pi^{-1}(r_1^{-1}\mathcal{P}^{[d_1]} \otimes^G r_2^{-1}\mathcal{P}^{[d_2]}) \cong \left(\bigotimes_{i=1}^{d_1}{}^{G} p_i^{-1}\mathcal{P}\right) \otimes^G \left(\bigotimes_{j=d_1+1}^{d_1+d_2}{}^{G} p_j^{-1}\mathcal{P}\right).$$

Since both pullbacks identify with

$$\bigotimes_{k=1}^{d_1+d_2}{}^{G} p_k^{-1}\mathcal{P}$$

in an $S_{d_1} \times S_{d_2}$-equivariant manner, the isomorphism descends to the quotient $X^{(d_1)} \times_S X^{(d_2)}$ by Proposition 19. $\square$

2.4 Algebraic Preliminaries on Ramification

This section reviews the ramification of discrete valuations, beginning with the general theory of *discrete valuation rings* and their integral closures in finite separable extensions. We then specialize to complete rings in Galois extensions to describe the ramification filtration via lower and upper numbering, following the treatments in [Stacks, Tag 0EXQ] and [Ser79].

Definition 25 ([Stacks, Tag 09E4])**.** We say that $A \to B$ or $A \subset B$ is an extension of discrete valuation rings if A and B are discrete valuation rings and $A \to B$ is injective and local. In particular, if π_A and π_B are uniformizers of A and B, then $\pi_A = u\pi_B^e$ for some $e \geq 1$ and unit u of B. The integer e does not depend on the choice of the uniformizers as it is also the unique integer ≥ 1 such that

$$\mathfrak{m}_A B = \mathfrak{m}_B^e$$

The integer e is called the *ramification index* of B over A. We say that B is *weakly unramified* over A if $e = 1$. If the extension of residue fields $\kappa_A = A/\mathfrak{m}_A \subset \kappa_B = B/\mathfrak{m}_B$ is finite, then we set $f = [\kappa_B : \kappa_A]$ and we call it the *residual degree* or residue degree of the extension $A \subset B$.

Note that we do not require the extension of fraction fields to be finite.

Now let A be a discrete valuation ring with fraction field K, let L/K be a finite separable field extension and let $B \subset L$ be the integral closure of A in L. Then the ring extension $A \subset B$ is finite, hence B is Noetherian. The dimension of B is 1, hence B is a Dedekind domain. Let $\mathfrak{m}_1, \dots, \mathfrak{m}_n$ be the maximal ideals of B (i.e., the primes lying over $\mathfrak{m}_A$). We obtain extensions of discrete valuation rings

$$A \subset B_{\mathfrak{m}_i}$$

and hence ramification indices e_i and residue degrees f_i. We have

$$[L : K] = \sum\nolimits_{i=1,\dots,n} e_i f_i$$

Note that if A is henselian (e.g. A is complete) then $n = 1$.

Definition 26 ([Stacks, Tag 09E9])**.** Let A be a discrete valuation ring with fraction field K. Let L/K be a finite separable extension. With B and $\mathfrak{m}_i$, $i = 1, \dots, n$ as above, we say the extension L/K is

1. unramified with respect to A if $e_i = 1$ and the extension $\kappa(\mathfrak{m}_i)/\kappa_A$ is separable for all i,
2. tamely ramified with respect to A if either the characteristic of κ_A is 0 or the characteristic of κ_A is $p > 0$, the field extensions $\kappa(\mathfrak{m}_i)/\kappa_A$ are separable, and the ramification indices e_i are prime to p, and
3. totally ramified with respect to A if $n = 1$ and the residue field extension $\kappa(\mathfrak{m}_1)/\kappa_A$ is trivial.

If the discrete valuation ring A is clear from context, then we sometimes say L/K is unramified, totally ramified, or tamely ramified for short.

Now and for the rest of the section assume A is a *complete discrete valuation ring* with uniformizer π and residue field κ, which we assume to be perfect. When A and κ are of the same characteristic

$p > 0$, then A contains a coefficient field $k \cong \kappa$ and a well known structure theorem holds: $A = k[[\pi]] \cong k[[t]]$. Let K be the fraction field of A, then $K = k((\pi))$.

For the remainder of this section, assume A is a *complete discrete valuation ring* with uniformizer π and a perfect residue field κ. In the equal characteristic case, where $\mathrm{char}(A) = \mathrm{char}(\kappa) = p > 0$, the Cohen Structure Theorem implies that A contains a coefficient field $k \cong \kappa$, such that $A \cong k[[\pi]] \cong k[[t]]$. Consequently, the fraction field is given by $K = k((\pi))$. Let L/K be a finite Galois extension with Galois group G. The integral closure B of A in L is itself a complete discrete valuation ring. Since the residue field κ is perfect, the extension of residue fields is separable, and there exists a uniformizer $\Pi \in B$ such that $B = A[\Pi]$.

The *lower numbering ramification filtration* of G, denoted by $(G_i)_{i \geq -1}$, is defined as:

$$G_i = \{\sigma \in G \mid v_B(\sigma(x) - x) \geq i + 1 \text{ for all } x \in B\}$$

where v_B is the valuation on L associated with B. In this indexing, $G_{-1} = G$ and G_0 corresponds to the *inertia group* of the extension L/K. Note that L/K is unramified if and only if $G_0 = \{1\}$, and it is tamely ramified if and only if $G_1 = \{1\}$.

A standard result in the theory of local fields(complete local fields with a discrete valuation and perfect residue field) ensures that the condition in the definition of G_i need only be checked for the uniformizer Π of B. Specifically, if we define the function

$$i_K^L(\sigma) = v_B(\sigma(\Pi) - \Pi)$$

for $\sigma \in G$, then $G_i = \{\sigma \in G \mid i_K^L(\sigma) \geq i + 1\}$. The subgroups G_i are normal in G and become trivial for sufficiently large i.

Consider a tower of fields $K \subset E \subset L$, and let $H = \mathrm{Gal}(L/E) \subset G$. The lower numbering is compatible with subgroups:

$$G_i \cap H = H_i \quad \text{for all } i \geq -1$$

which follows from the identity $i_E^L = i_K^L|_H$. However, the lower numbering does not behave as simply with respect to quotients. If H is normal in G, the image of G_i in G/H is generally not $(G/H)_i$. Instead, we have $G_i H/H = (G/H)_j$, where the index j is given by the formula:

$$j = \frac{1}{e_{L/E}} \sum_{\tau \in H} \min(i_K^L(\tau), i + 1) - 1$$

In the literature, one reindexes the ramification groups by defining the Herbrand function $\phi_{L/K} : [-1, \infty) \to [-1, \infty)$:

$$\phi_{L/K}(i) = \frac{1}{e_{L/K}} \sum_{\sigma \in G} \min(i_K^L(\sigma), i + 1) - 1 = \int_0^i \frac{1}{[G_0 : G_t]} dt$$

This function is continuous, strictly increasing, and piecewise linear, making it a bijection. It satisfies the composition law $\phi_{L/K} = \phi_{E/K} \circ \phi_{L/E}$ for a tower of extensions $K \subset E \subset L$. Using this function, we define the *upper numbering ramification groups* as:

$$G^i = G_{\phi_{L/K}^{-1}(i)}$$

The primary advantage of this reindexing is its compatibility with quotients: for any normal subgroup $H \subset G$, we have:

$$G^i H/H = (G/H)^i$$

for all $i \geq -1$.

Proposition 27 ([Ser79], IV Corollary 3). *the quotients G_i/G_{i+1}, $i \geq 1$, are abelian groups, and are direct products of cyclic groups of order p. The group G_1 is a p-group.*

Proposition 28 ([Ser79], IV Corollary 4). *The inertia group G_0 has the following property: It is the semi-direct product of a cyclic group of order prime to p with a normal subgroup whose order is a power of p.*

Kummer and Artin-Schreier Theories

For cyclic extensions, Kummer and Artin-Schreier theories provide explicit ways to calculate ramification. These results show how the valuation of a defining element $a \in K$ determines the ramification index and the jumps in the upper numbering filtration. Throughout this section, let K be a discrete valuation field with a perfect residue field κ of characteristic $p > 0$.

Theorem 29 (Ramification in Kummer Extensions, [Koc97], Proposition 1.83). *Assume K contains the n-th roots of unity μ_n. Let L/K be the extension given by the equation $X^n = a$ for some $a \in K^\times$ and denote by G its Galois group. Then we have:*

1. *If $v_K(a) \in n\mathbb{Z}$ and the image of $a\pi^{-v_K(a)}$ in the residue field κ is an n-th power, the extension L/K is trivial.*

2. *If $v_K(a) \in n\mathbb{Z}$ and the image of $a\pi^{-v_K(a)}$ in the residue field κ is not an n-th power, the extension L/K is cyclic and unramified.*

3. *If $v_K(a) \notin n\mathbb{Z}$, the extension L/K is cyclic and ramified. Specifically, if $\gcd(|v_K(a)|, n) = 1$, the extension is totally ramified of degree n. Otherwise it has ramification index $\frac{n}{|gcd(v_K(a)|,n)}$*

Conversly, Kummer theory ensures that every cyclic extension of degree n, prime to p of a field that contains n-th roots of unity, is of the above form. Moreover, in the above we can always take $a \in \mathcal{O}_K$

Note that in the case of total ramification the extesntion is tamely ramified.

Theorem 30 (Ramification in Artin-Schreier Extensions, [Tho05]). *Let $\wp(x) = x^p - x$ be the Artin-Schreier opeartor. Let L/K be the extension given by the equation $X^p - X = a$ for some $a \in K$ and denote by G its Galois group. Then we have:*

1. *If $v_K(a) > 0$ or if $v_K(a) = 0$ and $a \in \wp(K)$, the extension L/K is trivial.*

2. *If $v_K(a) = 0$ and if $a \notin \wp(K)$, the extension L/K is cyclic of degree p and unramified.*

3. *If $v_K(a) = -m < 0$ with $m \in \mathbb{Z}_{>0}$ and if m is prime to p, the extension L/K is cyclic of degree p again and totally ramified. Moreover, its ramification groups are given by:*

$$G = G^{(-1)} = \cdots = G^{(m)} \quad \text{and} \quad G^{(m+1)} = 1.$$

Conversely, Artin-Schreier theory ensures that every cyclic extension of degree p takes this form. Moreover, in the above and under the isomorphism $K \cong k((t))$, one can always take a of the form $ct^{-m} + a_{-m+1}t^{-m+1} + \cdots a_{-1}t^{-1} + a_0$. If k is algebraically closed then there is a change of variables such that $a = u^{-m}$.

2.5 Algebraic Geometry

In this section we group together some general theorems in algebraic geometry that we will be employing throughtout the text. All schemes are assumed to be locally of finite type.

Theorem 31. *Let $f : X \to Y$ be a finite flat map between integral schemes, of finite type over a field k. if $Z \subset X$ is a prime divisor with generic point η_Z, then $f(Z) \subset X$ is a prime divisor with generic point $\eta_{f(Z)}$ satisfying $f(\eta_Z) = \eta_{f(Z)}$*

Proof. f is finite hence proper hence closed so $f(Z)$ is closed subset of Y, it is irreducible as the image of an irreducible. Since $Z = \overline{\{\eta_Z\}}$ we get:

$$\{f(\eta_Z)\} \subset f(Z) = f(\overline{\{\eta_Z\}}) \subseteq \overline{f(\{\eta_Z\})} = \overline{\{f(\eta_Z)\}}$$

And since $f(Z)$ is closed we get $f(Z) = \overline{\{f(\eta_Z)\}}$.

For flat map of integral schemes we have for every $x \in X$, $y = f(x)$ the dimension formula:

$$\dim(\mathcal{O}_{X,x}) = \dim(\mathcal{O}_{Y,y}) + \dim(\mathcal{O}_{X_y,x})$$

And since $\dim(\mathcal{O}_{X_y,x}) = 0$ we get $\dim(\mathcal{O}_{X,x}) = \dim(\mathcal{O}_{Y,y})$ concluding that $f(Z)$ is a prime divisor as well. □

A known theorem states that:

Theorem 32. *Let X, Y be two integral schemes over a field k. If X is geometrically integral then $X \times_k Y$ is integral. If both X, Y are geometrically integral, then $X \times_k Y$ is geometrically integral.*

Theorem 33. *Let C be smooth projective curve geometrically connected over a field k. Then:*

1. *$C^{(d)}$ is smooth*
2. *For every d, $C^{(d)}$ is integral.*
3. *For every d, $C^{(d)}$ is geometrically integral.*
4. *The product of every finite number of $C^{(d)}$ is geometrically integral.*

Proof.

1. Let t_i be a local parameter for C at P_i. The local ring of the product C^d at the point $(P_1, \ldots, P_d)$ is isomorphic $k[[t_1, t_2, \ldots, t_d]]$ and the local ring of the qoutient at the divisor $D = \sum P_i$ is $k[[t_1, \ldots, t_d]]^{S_d}$ which is isomorphic to $k[[t_1, \ldots, t_d]]^{S_d} \cong k[[s_1, \ldots, s_d]]$ where the s_i are the symmetric polynomials, hence this ring is regular local ring.
2. C is irreducible hence C^d is irreducible hence $C^{(d)}$ is irreducible. Since $C^{(d)}$ is smooth it is reduced.
3. By [Stacks, Tag 0366], C is geometrically integral, so it follows from the above.
4. Theorem 32

□

Blowups

Theorem 34 ([Stacks, Tag 0805]). *Let $X_1 \to X_2$ be a flat morphism of schemes. Let $Z_2 \subset X_2$ be a closed subscheme. Let Z_1 be the inverse image of Z_2 in X_1. Let X_i' be the blowup of Z_i in X_i. Then there exists a cartesian diagram*

$$\begin{tikzcd} X_1' \arrow[r] \arrow[d] & X_2' \arrow[d] \\ X_1 \arrow[r] & X_2 \end{tikzcd}$$

of schemes.

Theorem 35. *If X is integral then $Bl_Z(X)$ is integral.*

If X is a smooth curve, we have the following result regarding its symmetric power:

Proposition 36. *Let $f : X \to S$ be a smooth morphism of relative dimension 1. Suppose f is flat and Zariski-locally quasi-projective. Then the relative symmetric power $X_S^{(d)} = Sym_S^d(X)$ is smooth over S of relative dimension d.*

Proof. Since $f : X \to S$ is smooth, the d-fold product X_S^d is smooth over S. By Proposition 19, the quotient $X_S^{(d)}$ is flat over S. Smoothness is a fiberwise property for flat morphisms of finite presentation (cf. [Stacks, Tag 01V8]); thus, it suffices to verify the smoothness of the fibers. For every $s \in S$, the fiber of the symmetric power is given by:

$$(\mathrm{Sym}_S^d(X))_s \cong \mathrm{Sym}_{\kappa(s)}^d(X_s)$$

Consequently, we may reduce to the case where $S = \mathrm{Spec}(k)$ for a field k. Since smoothness is preserved under base change to the algebraic closure, we may further assume $k = \bar{k}$. Let $z = \sum_{i=1}^r d_i \cdot x_i \in X^{(d)}$ be a point represented by an effective cycle of degree d, where the points $x_i \in X(k)$ are distinct and $\sum d_i = d$. Choose pairwise disjoint Zariski-open subsets $U_i \subset X$ such that $x_i \in U_i$. Let $W \subset X^{(d)}$ be the open subset consisting of cycles whose support is contained in $\bigcup U_i$ and which meet each U_i with degree exactly d_i. There is a canonical isomorphism:

$$W \cong \prod_{i=1}^r U_i^{(d_i)}$$

Thus, it suffices to show that each $U_i^{(d_i)}$ is smooth of dimension d_i at the point $d_i \cdot x_i$.

We may therefore restrict our attention to the "worst" case: the diagonal point $z = d \cdot x$. Let $R = \mathcal{O}_{X,x}$ be the local ring of the curve at x. Since X is a smooth curve over k, R is a regular local ring of dimension 1, i.e., a discrete valuation ring. Its completion is $\hat{R} \cong k[[t]]$. The completed local ring of the product X^d at the point $(x, \dots, x)$ is:

$$\hat{\mathcal{O}}_{X^d,(x,\dots,x)} \cong \hat{R} \hat{\otimes}_k \dots \hat{\otimes}_k \hat{R} \cong k[[t_1, \dots, t_d]]$$

where t_i is the uniformizer for the i-th copy of X. The symmetric group S_d acts on this power series ring by permuting the variables t_i. By the *Fundamental Theorem of Symmetric Polynomials*, the ring of invariants is:

$$\left(k[[t_1, \dots, t_d]]\right)^{S^d} = k[[e_1, \dots, e_d]]$$

where e_j is the j-th elementary symmetric power series in the variables t_i.

The ring $k[[e_1, \ldots, e_d]]$ is a formal power series ring in d variables. A fundamental theorem in commutative algebra states that a local ring is regular if and only if its completion is a formal power series ring over a field. Since $\hat{\mathcal{O}}_{X^{(d)},z} \cong k[[e_1, \ldots, e_d]]$, the local ring $\mathcal{O}_{X^{(d)},z}$ is a regular local ring of dimension d. This implies that $X^{(d)}$ is smooth over k of dimension d, completing the proof. $\square$

3 Ramification After Blowup Is Tame

Throughout this section let C be a projective smooth geometrically connected curve over k. Let $\mathfrak{m} = \sum n_i P_i$ be a fixed modulus. Let $U = C \setminus \mathfrak{m}$. For any modulus $\mathfrak{n} \subset \mathfrak{m}$, we define $Z_{\mathfrak{n}}$ as the closed subscheme of $C^{(\deg \mathfrak{n})} = \mathrm{Sym}_k^{\deg \mathfrak{n}}(C)$ defined by $\mathfrak{n}$ as a point of $C^{(\deg \mathfrak{n})}$ We then define $X_{\mathfrak{n}}$ as the blowup of $C^{(\deg \mathfrak{n})}$ at $Z_{\mathfrak{n}}$, and we denote by $E_{\mathfrak{n}} = Z_{\mathfrak{n}} \times_{C^{(d)}} X_{\mathfrak{n}}$ the exceptional divisor of this blowup, it is irreducible of codimension 1. We denote by $\eta_{\mathfrak{n}}$ the generic point of $E_{\mathfrak{n}}$.

Diagrammatically:

$$\begin{tikzcd} \overline{\{\eta_{\mathfrak{n}}\}} = E_{\mathfrak{n}} \arrow[r, hook] \arrow[d] & X_{\mathfrak{n}} \arrow[d, "\pi_{\mathfrak{n}}"] \\ Z_{\mathfrak{n}} \arrow[r, hook] & C^{(\deg \mathfrak{n})} \end{tikzcd} \tag{4}$$

Our main goal in this section is to prove Theorem 3:

Theorem 37. *Let G be a finite abelian group. And let $\mathcal{P} \to U$ be a G-torsor in $U_{\acute{E}t}$ with ramification bounded by $\mathfrak{m} = \sum n_i P_i$. Let $\mathfrak{n} = n_i P_i \subset \mathfrak{m}$ be a sub modulus of $\mathfrak{m}$. Then $\mathcal{P}^{[\deg \mathfrak{n}]}$ is tamely ramified at $\eta_{\mathfrak{n}}$.*

Our approach proceeds in three stages. First, we establish the result for the cyclic case $G = \mathbb{Z}/p\mathbb{Z}$ and for groups G where $\gcd(|G|, p) = 1$. Next, we extend the proof to cyclic p-groups $G = \mathbb{Z}/p^r\mathbb{Z}$. Finally, for a general finite abelian group G, we conclude by applying the structure theorem for finite abelian groups and Proposition 14 to decompose the torsor into these constituent cases.

3.1 Ramification of G-Torsors

Generally speaking, assume we are given a locally Noetherian scheme X, a dense open $U \subset X$, a finite étale morphsm $f : Y \to U$ and a prime divisor $Z \subset X$ such that $Z \cap U = \emptyset$ and the local ring $\mathcal{O}_{X,\xi}$ of X at the generic point ξ of Z is a discrete valuation ring. Setting $K_\xi = \mathrm{Frac}(\mathcal{O}_{X,\xi})$ we obtain a cartesian square

$$\begin{tikzcd} \mathrm{Spec}(K_\xi) \arrow[r] \arrow[d] & U \arrow[d] \\ \mathrm{Spec}(\mathcal{O}_{X,\xi}) \arrow[r] & X \end{tikzcd}$$

of schemes. In particular, we see that $Y \times_U \mathrm{Spec}(K_\xi)$ is the spectrum of a finite separable algebra L_ξ / K_ξ.

Definition 38. We say:

1. Y is *unramified* over X at (Z, ξ) resp. Y is *tamely ramified* over X at (Z, ξ) if L_ξ / K_ξ is unramified, resp. tamely ramified with respect to $\mathcal{O}_{X,\xi}$ (Definition 26)

2. Y is unramified over X in codimension 1, resp. Y is tamely ramified over X in codimension 1 if L_ξ/K_ξ is unramified, resp. tamely ramified with respect to $\mathcal{O}_{X,\xi}$ for every (Z,ξ) as above.

More precisely, we decompose L_ξ into a product of finite separable field extensions of K_ξ and we require each of these to be unramified, resp. tamely ramified with respect to $\mathcal{O}_{X,\xi}$.

Now back to our orginial assumptions, C is projective smooth geometrically connected curve over a field k and $U = C \setminus \mathfrak{m}$. Every (Z,ξ) is now a closed point P and $\mathcal{O}_{C,P}$ is a discrete valuation ring. Assume $f : Y \to U$ is actually G-torsor $\mathcal{P} \to U$ in $U_{\text{Ét}}$ for G finite abelian group.

By passing to the completion $\widehat{\mathcal{O}}_{C,P}$, we obtain the complete local field $\widehat{K}_P$. Restricting the torsor $\mathcal{P}$ to the punctured formal neighborhood $\mathrm{Spec}(\widehat{K}_P)$ allows us to study the ramification in a strictly local context. This restriction yields a G-torsor over a field, which necessarily decomposes into a disjoint union of spectra of finite separable field extensions

$$\mathcal{P}|_{\mathrm{Spec}(\widehat{K}_P)} \cong \bigsqcup \mathrm{Spec}(F)$$

such that:

1. These extensions $F/\widehat{K}_P$ are all isomorphic and Galois.
2. The Galois group $H = \mathrm{Gal}(F/\widehat{K}_P)$ identifies with the stabilizer of any connected component of the pullback $\mathcal{P}|_{\mathrm{Spec}(\widehat{K}_P)}$, which is a subgroup of G.
3. The number of such components is then given by the index $[G : H]$.

Definition 39. We say that the extension $F/\widehat{K}_P$ *has ramification bounded by* r if the ramification group H^r is trivial. We say $\mathcal{P}$ *has ramification bounded by* r *at* P if the corresponding local field extensions $F/\widehat{K}_P$ satisfy this condition.

Definition 40. Let $\mathfrak{m} = \sum n_i P_i$ be a modulus on C. We say the G-torsor $\mathcal{P} \to U$ has **ramification bounded by** $\mathfrak{m}$ if, for each point P_i, the local restriction $\mathcal{P}|_{\mathrm{Spec}(\widehat{K}_{P_i})}$ has ramification bounded by the coefficient n_i.

This local property remains invariant under base change to the separable closure. Let $\bar{s} = \mathrm{Spec}(\bar{k}) \to \mathrm{Spec}(k)$ be a geometric point. The higher ramification groups for $\mathcal{P}|_{\mathrm{Spec}(\widehat{K}_P)}$ are isomorphic to those of the geometric restriction $\mathcal{P}_{\bar{k}}$ at any point $\bar{x}$ lying over P. Thus an equivalent defintion:

Definition 41. Let $\mathfrak{m} = \sum n_i P_i$ be a modulus on C. The G-torsor $\mathcal{P} \to U$ has **ramification bounded by** $\mathfrak{m}$ if for every geometric point $\bar{x}$ of $\mathfrak{m}$ (with image $\bar{s}$ in $\mathrm{Spec}(k)$), the restriction of $\mathcal{P}$ to

$$\mathrm{Spec}(\widehat{\mathcal{O}}_{C_{\bar{k}},\bar{x}}) \times_{C_{\bar{k}}} U_{\bar{k}}$$

has ramification bounded by the multiplicity of $\mathfrak{m}_{\bar{s}}$ at $\bar{x}$ in the sense of Definition 39.

These definitions are equivalent because $\widehat{\mathcal{O}}_{C_{\bar{k}},\bar{x}} \cong \widehat{\mathcal{O}}_{C,P} \otimes_k \bar{k}$. Furthermore, the notions of tame ramification and unramifiedness derived here coincide with the codimension-1 criteria in Definition 38.

There is another equiviliant formulation through the language of characters. The group G, being a finite abelian group over k, is étale. Consequently, the correspondence between G-torsors and the fundamental group allows us to describe G-torsors via characters. Specifically, there is an isomorphism: (Corollary 13)

$$\mathrm{Hom}_{\mathrm{cont.}}(\pi_1^{\text{ét}}(U,\overline{x}),G) \xrightarrow{\cong} \mathrm{Tors}(U_{\text{ét}},G)$$

In our local setting, where we consider the restriction to the complete local field $\widehat{K}_P$, the fundamental group identifies with the absolute Galois group $G_{\widehat{K}_P} := \mathrm{Gal}(\widehat{K}_P^{\mathrm{sep}}/\widehat{K}_P)$. Thus, the restricted G-torsor $\mathcal{P}|_{\mathrm{Spec}(\widehat{K}_P)}$ corresponds to a unique continuous homomorphism:

$$\rho : G_{\widehat{K}_P} \to G$$

Under this correspondence, the torsor $\mathcal{P}$ has ramification bounded by r at P if and only if the homomorphism ρ trivializes the r-th ramification group in the upper numbering:

$$\rho(G^r_{\widehat{K}_P}) = \{1\}$$

Basic Properties of Ramification of G-Torsors

We now prove some elementary lemmas regarding the ramification of G-torsors.

Lemma 42. *Let G be a finite abelian group and X be a locally Noetherian scheme over a field k. Let $U \subset X$ be a dense open subset and let (Z,ξ) be a prime divisor in the complement $X \setminus U$. Assume $\mathcal{P}_1$ and $\mathcal{P}_2$ are two G-torsors on $U_{ét}$, Such that $\mathcal{P}_1$ has ramification bounded by r_1 at (Z,ξ), and $\mathcal{P}_2$ has ramification bounded by r_2 at (Z,ξ). Then their product $\mathcal{P}_1 \otimes^G \mathcal{P}_2$ has ramification bounded by $max(r_1,r_2)$ at (Z,ξ).*

Proof. Let $A = \widehat{\mathcal{O}_{X,\xi}}$ and let $K = Frac(A)$. Let $\rho_1,\rho_2 : G_K \to G$ be the associated continious homomorphisms corresponding to the G-torsors $\mathcal{P}_1|_{\mathrm{Spec}(K)}$, $\mathcal{P}_2|_{\mathrm{Spec}(K)}$. We finish by noticing that the associated character of $(\mathcal{P}_1 \otimes^G \mathcal{P}_2)|_{\mathrm{Spec}(K)}$ is $\rho = \rho_1 + \rho_2$. ☐

Lemma 43. *Let $f : X \to Y$ be a morphism of locally Noetherian schemes. Let $U_X \subset X$ and $U_Y \subset Y$ be dense open subschemes such that $f^{-1}(U_Y) \subset U_X$.*

Let (Z_X,ξ_X) and (Z_Y,ξ_Y) be prime divisors of X and Y such that $Z_X \cap U_X = \emptyset$ and $Z_Y \cap U_Y = \emptyset$. Suppose $f(\xi_X) = \xi_Y$ and that f is étale at ξ_X. Let $\mathcal{P}$ be a G-torsor over U_Y, and let $f^{-1}\mathcal{P}$ be its pullback to $f^{-1}(U_Y) \subset U_X$. Then, the ramification of $f^{-1}\mathcal{P}$ at ξ_X is bounded by r if and only if the ramification of $\mathcal{P}$ at ξ_Y is bounded by r.

Proof. The boundedness of ramification is determined by the behavior of the torsor over the completion of the local rings at the generic points. Let $A = \mathcal{O}_{Y,\xi}$ and $B = \mathcal{O}_{X,\xi_X}$ be the discrete valuation rings at the generic points, with fraction fields K and L respectively. Since f is étale at ξ_X, the map $A \to B$ is a flat, unramified (hence weakly unramified) local homomorphism. Consequently, the extension of completions $\widehat{L}/\widehat{K}$ is a finite unramified extension of complete discretely valued fields. We finish by noticing that the upper numbering filtration on the absolute Galois group is compatible with unramified base change.[5] ☐

[5]Specifically, let $G_K = \mathrm{Gal}(K^{sep}/K)$ and $G_L = \mathrm{Gal}(L^{sep}/L)$. For an unramified extension, the Herbrand function is the identity, which implies that for any $r \geq 0$:

$$G^r_L = G^r_K \cap G_L$$

Let C be a smooth, proper (projective) curve over a field k, and let $U \subset C$ be a dense open subset. Let G be a finite abelian group. and let $\pi_U : U' \to U$ be a G-torsor in $U_{\text{Ét}}$. Let C' be the normal proper model of U' over k (i.e. the unique smooth proper curve which has U' as a dense open subset). Let $\pi : C' \to C$ be the induced morphism. (which is finite generically étale over U). Let $U'^{(d)}$ be the standard d-th symmetric product of U' as a scheme, and let $U'^{[d]}$ be the torsor-theoetric symmetric product, by Corollary 22 we have a canonical morphism $q_U : U'^{(d)} \to U'^{[d]}$. Note that $C'^{(d)}$ is smooth and finite over $C^{(d)}$, hence it is the normalization of $C^{(d)}$ inside $K(C'^{(d)})$. $U'^{(d)}$ is a dense open subset of $C'^{(d)}$, hence $C'^{(d)}$ is a normal proper model of $U'^{(d)}$. Let $C'^{[d]}$ be the normalization of $C^{(d)}$ in the function field $K(U'^{[d]})$, then q_U extends to a map $q : C'^{(d)} \to C'^{[d]}$, and we have the following diagrams:

$$\begin{tikzcd} U' \arrow[r, hook] \arrow[d, "\pi_U"'] & C' \arrow[d, "\pi"] \\ U \arrow[r, hook] & C \end{tikzcd} \qquad \begin{tikzcd} U'^{(d)} \arrow[r, hook] \arrow[d, "q_U"'] & C'^{(d)} \arrow[d, "q"] \\ U'^{[d]} \arrow[r, hook] \arrow[d] & C'^{[d]} \arrow[d] \\ U^{(d)} \arrow[r, hook] & C^{(d)} \end{tikzcd}$$

Lemma 44. *In the context of the preceding discussion, the morphism $U'^{(d)} \to U'^{[d]} \hookrightarrow C'^{[d]}$ is unramified in codimension* 1 *(see Definition 38).*

Proof. Let $C' \setminus U' = \{x_1, \ldots, x_m\}$. For each $x \in C' \setminus U'$, define the locus

$$Z_x = \{D \in C'^{(d)} \mid x \in \operatorname{Supp}(D)\}.$$

Each Z_x is a prime divisor in the symmetric product $C'^{(d)}$, and the boundary $C'^{(d)} \setminus U'^{(d)}$ is precisely the finite union $\bigcup_{i=1}^{m} Z_{x_i}$.

Let η be the generic point of Z_x. It suffices to show that the local ring extension $\mathcal{O}_{C'^{[d]},q(\eta)} \hookrightarrow \mathcal{O}_{C'^{(d)},\eta}$ is weakly unramified (Definition 25).

Let $p : C'^d \to C'^{(d)}$ be the natural quotient. Consider the closed subvariety

$$H = \{x\} \times C' \times \cdots \times C' \subset C'^d.$$

The subvariety H is irreducible, and its image under p is exactly Z_x. Consequently, p maps the generic point η_H of H to the generic point η of Z_x.

$$\begin{tikzcd} \eta_H \arrow[r] \arrow[d] & H \arrow[r] \arrow[d] & C'^d \arrow[d, "p"] \\ \eta \arrow[r] \arrow[d] & Z_x \arrow[r] \arrow[d] & C^{(d)} \arrow[d, "q"] \\ q(\eta) \arrow[r] & q(Z_x) \arrow[r] & C'^{[d]} \end{tikzcd}$$

It suffices to show that the local ring extension $\mathcal{O}_{C'^{[d]},q(\eta)} \hookrightarrow \mathcal{O}_{C'^d,\eta_H}$ is weakly unramified. Let $R = \mathcal{O}_{C'^d,\eta_H}$. And let $A = \mathcal{O}_{C'^{[d]},q(\eta)}$.

Recall that $C'^{[d]}$ is the quotient of C'^d by the finite group $\Xi = K_G \rtimes S_d$ (see Corollary 22), and the morphism $q \circ p : C'^d \to C'^{[d]}$ is *branched Galois Cover* (finite, surjective map of normal integral schemes $X \to Y$, with function field extension $K(X)/K(Y)$ Galois).

Hence the decomoposition group of η_H in Ξ and the inertia group of η_H in Ξ are defined:

$$D_{\eta_H} = \{g \in \Xi \mid g(\eta_H) = \eta_H\}$$

$$I_{\eta_H} = \{g \in D_{\eta_H} \mid g^*(u) \equiv u \pmod{\mathfrak{m}_R} \text{ for all } u \in R\}$$

Claim: The inertia group I_{η_H} is trivial.

Let $K = K(C')$ be the function field of C', then $K(C'^d) = K^{\otimes_k d} \cong Frac(K_1 \otimes_k K_2 \otimes_k \cdots \otimes_k K_d)$ Where K_i is the copy of K corresponding to the i-th factor of C'^d. The closed subscheme H corresponds to the prime ideal $\mathfrak{p}_H \subset \mathcal{O}_{C'^d}$, and $R = \mathcal{O}_{C'^d, \mathfrak{p}_H}$. The residue field of R is exactly $K(H) \cong Frac(k(x) \otimes_k K_2 \otimes_k \cdots \otimes_k K_d)$ where $k(x)$ is the residue field of x.

The action of $\gamma = (g_1, \ldots, g_d, \sigma) \in \Xi$, via the pullback (γ^*) on $K(C'^d)$ does two things (in that order):

1. It permutes the tensor factors according to σ. A function strictly in K_i is mapped to a function in $K_{\sigma(i)}$.
2. It applies the field automorphism $g_i^* \in \mathrm{Aut}(K/k)$ to the respective factors.

Let $\gamma \in I_{\eta_H} \subset \Xi$, Then the induced automorphism $\bar{\gamma}^*$ on $K(H)$ must be the identity map. This implies that σ must fix i for every $i \geq 2$ (The K_i are linearly independent), and hence $\sigma = id$. The trivial action of $\bar{\gamma}^*$ restricts to a trivial action of g_i^* on K_i for every $i \geq 2$, and hence $g_i = 1$ for every $i \geq 2$ (since the action of G on $K(C')$ is faithful). Finally, since $g_1 g_2 \cdots g_d = 1$, we also have $g_1 = 1$. Since the inertia group I_{η_H} is trivial, the local ring extension $A \hookrightarrow R$ is weakly unramified. (See for example [Stacks, Tag 09E3], [Stacks, Tag 0BTF])

□

3.2 Proof of Theorem 3

We argue its enough to assume $C = \mathbb{P}^1_k$ and $\mathcal{P} \to \mathbb{G}_m \subset \mathbb{P}^1_k$ is our G-torsor. This is essentially by employing the following principle:

Invaraince of Local Invariants under Formal isomorphism: *If two schemes have isomorphic completed local rings at given points, then any construction obtained by functorial algebraic operations (products, quotients by finite groups, blowups, finite covers) produces isomorphic completed local rings at the corresponding points, and hence identical ramification behavior.*

Which applies for all of our constructions- blowups, symmetric products, etc. So in what follows, we make our life simpler by assuming the above. So, let G be a finite abelian group and assume $C = \mathbb{P}^1_k$, $\mathfrak{m} = d \cdot 0$ and $\mathbb{G}_m = U \subset U' = C \setminus \mathfrak{m}$. Then $\deg \mathfrak{m} = d$ We also assume k is algebraicly closed. (We can etale base change, and this doesn't change ramification.)

We start by analyzing the blowup $\tilde{X} = \mathrm{Bl}_0(X)$ of $X = \mathbb{A}^n_k$ where $0 \in X$ be the origin. Recall that $\tilde{X} \subset X \times_k \mathbb{P}^{n-1}_k$ is defined by the equations $x_i u_j = x_j u_i$, where $[u_1 : \cdots : u_n]$ are the homogeneous coordinates of $\mathbb{P}^{n-1}_k$. The exceptional divisor $E \subset \tilde{X}$ is the fiber over the origin,

$E = \{(0, [u_1 : \cdots : u_n])\}$, which is of codimension 1 in $\tilde{X}$. Let $\eta \in E$ be the generic point of E, and let $R = \mathcal{O}_{\tilde{X},\eta}$ be the associated local ring. This ring R is a discrete valuation ring (DVR) with fraction field $K = K(X) = k(x_1, \dots, x_n)$.

On the affine chart U_1 where $u_1 \neq 0$, we have $x_i = \frac{u_i}{u_1} x_1$. The coordinate ring is:

$$\mathcal{O}_{\tilde{X}}(U_1) = k\left[x_1, \frac{u_2}{u_1}, \dots, \frac{u_n}{u_1}\right]$$

In this chart, the generic point η corresponds to the prime ideal $\mathfrak{p}_1 = (x_1)$. Thus, the local ring is $R = k[x_1, \frac{u_2}{u_1}, \dots, \frac{u_n}{u_1}]_{(x_1)}$. The residue field is $\kappa(\eta) = k(\frac{u_2}{u_1}, \dots, \frac{u_n}{u_1})$, and the completion of R with respect to its maximal ideal is:

$$\hat{R} = \kappa(\eta)[[x_1]]$$

In this local ring, x_1 is a uniformizer. Note that any x_i (for $i > 1$) can also serve as a uniformizer, as $x_i = (\frac{u_i}{u_1})x_1$ and $\frac{u_i}{u_1}$ is a unit in R.

For any monomial $M = x_1^{a_1} \dots x_n^{a_n} \in K$, we can write:

$$M = x_1^{\sum a_i} \left(\frac{u_2}{u_1}\right)^{a_2} \dots \left(\frac{u_n}{u_1}\right)^{a_n}$$

Since the term in parentheses is a unit in R, the valuation ν_E associated with E satisfies:

$$\nu_E(M) = \sum a_i = \deg M$$

Consequently, for any polynomial $f = f_d + f_{d+1} + \cdots + f_l$, where f_i is the homogeneous part of degree i, we have $\nu_E(f) = d$ (the order of vanishing at the origin).

Our first result is:

Theorem 45. *Let $\mathcal{P} \to \mathbb{G}_m \subset \mathbb{P}^1_k$ be a G torsor which is either*

1. *tamely ramified at* 0
2. *totally ramified at* 0 *with $G = \mathbb{Z}/p\mathbb{Z}$ and ramificaiton bounded by d.*

Then the ramification of the G-torsor $\mathcal{P}^{[d]} \to \mathbb{G}_m^{(d)}$ at $\eta_{\mathfrak{m}}$ the generic point of $E_{\mathfrak{m}} \subset X_{\mathfrak{m}}$ is

1. *tamely ramified if $\mathcal{P}$ was tamely ramified*
2. *unramified if $\mathcal{P}$ was totally ramified with $G = \mathbb{Z}/p\mathbb{Z}$ and ramification bounded by d*

Proof. The local ring at the generic point of the excptional divisor of the blowup of the affine space at 0 point is $R = k[x_d, \frac{u_1}{u_d}, \dots, \frac{u_{d-1}}{u_d}]_{(x_d)}$. Where for every $i < d$, we have $x_i = \frac{u_i}{u_d} x_d$ are all uniformizers. The residue field is $\kappa(\eta) = k(\frac{u_1}{u_d}, \dots, \frac{u_{d-1}}{u_d})$ and the completion of R with respect to its maximal ideal is:

$$\hat{R} = \kappa(\eta)[[x_d]]$$

In our situation, when we take symmetric product of the affine space, the situation is similiar with different coordinates if we let $e_1, \dots, e_d$ be the symmetric polynomials in $x_1, \dots x_d$ then: The local ring is $R = k[e_d, \frac{u_2}{u_d}, \dots, \frac{u_{d-1}}{u_d}]_{(e_d)}$. $e_i = \frac{u_i}{u_d} e_d$ are all uniformizers. The residue field being:

$\kappa(\eta) = k(\frac{u_1}{u_d}, \dots, \frac{u_{d-1}}{u_d})$ and the completion: $\hat{R} = \kappa(\eta)[[s_d]]$ Note that from $e_i = \frac{u_i}{u_d} e_d$ We get $\frac{e_i}{e_d} = \frac{u_i}{u_d}$ in the fracion field. hence

$$\hat{K} = k(\frac{e_1}{e_d}, \dots, \frac{e_{d-1}}{u_d})((e_d)) \tag{5}$$

We compute directly the extesntion of complete valued fields over the complete valued field at the generic point. Note that by Theorem 29 and Theorem 30 We can assume $\mathcal{P} = \operatorname{Spec} k[x, x^{-1}][X]/(X^n - a)$ for $a \in k[x, x^{-1}]$ or $\mathcal{P} = \operatorname{Spec} k[x, x^{-1}][X]/(X^p + X - f(x, x^{-1}))$ where $f(x, x^{-1}) = cx^{-m} + a_{-m+1}x^{-m+1} + \cdots a_{-1}x^{-1} + a_0 = cx^{-m} + f_{-m+1}(x^{-1})$ where $m < d$.

We deal with each case separately.

Artin-Schreier Extensions:

Set $R = k[x, x^{-1}]$, and $S = \operatorname{Spec} k[x, x^{-1}][X]/(X^p + X - f(x^{-1}))$ we have

$$\begin{aligned}
R^{\otimes_k d} &= k[x_1, x_1^{-1}, \dots, x_d, x_d^{-1}] \\
S^{\otimes_k d} &= k[x_1, x_1^{-1}, \dots, x_d, x_d^{-1}][X_1, \dots, X_d]/(X_1^p - X_1 - f(x_1), \dots, X_d^p - X_d - f(x_d)) \\
&= R^{\otimes_k d}[X_1, \dots, X_d]/(X_1^p - X_1 - f(x_1), \dots, X_d^p - X_d - f(x_d))
\end{aligned}$$

Next, we want to understand the ring corresponding to $p_1^{-1}(\mathcal{P}) \otimes \cdots \otimes p_d^{-1}(\mathcal{P})$ on C^d - the d'th-product of the $G = \mathbb{Z}/p\mathbb{Z}$-torsors $p_1^{-1}(\mathcal{P}), \dots, p_d^{-1}(\mathcal{P})$ on U^d. It corresponds to qoutient: $\left(p_1^{-1}(\mathcal{P}) \times \cdots \times p_d^{-1}(\mathcal{P})\right)/G^{d-1}$ where the action of G^{d-1} on the product is:

$$(g_1, \dots, g_{d-1}) \cdot (p_1, p_2, \dots, p_{d-1}, p_d) = (g_1(p_1), g_1^{-1}g_2(p_2), \dots, g_{d-2}^{-1}g_{d-1}(p_{d-1}), g_{d-1}^{-1}(p_d))$$

The affine ring corrsponding to the torsor product is $\left(S^{\otimes_k d}\right)^{G^{d-1}}$

Recall that the action of $g \in G = \mathbb{Z}/p\mathbb{Z}$ on X is $g(X) = X + g$ (g correspond to a number $0 \le g \le p-1$). So, the action of $(g_1, \dots, g_{d-1})$ on the generators $(X_1, X_2, \dots, X_{d-1}, X_d)$ Is $X_1 \mapsto X_1 + g_1$, $X_i \mapsto X_i - g_{i-1} + g_i$ for $1 < i < d$ and $X_d \mapsto X_d - g_{d-1}$. So we see that $Y = X_1 + \cdots + X_d$ is invariant. Moreover $Y^P - Y - \sum_{i=1}^d f(x_i) = 0$ is irreducible degree p equation for Y, Since we are quotienting a rank p^d extesntion by a group of order p^{d-1} the resulting invaraint subring must have rank p over $R^{\otimes_k d}$, So we conclude:

$$\left(S^{\otimes_k d}\right)^{G^{d-1}} \cong R^{\otimes_k d}[Y]/(Y^p - Y - \sum_{i=1}^d f(x_i))$$

The group S_d acts on $R^{\otimes_k d} = k[x_1^{\pm 1}, x_2^{\pm 1}, \dots, x_d^{\pm 1}]$ by permuting the variables $\{x_i\}_{i=1}^d$. And since $Y = \sum_i^d X_i$ it leaves Y invaraint. The invaraint subring $(R^{\otimes_k d})^{S_d}$ is simply $k[e_1, e_2, \dots, e_d, e_d^{-1}]$ where $\{e_i\}$ are the symmetric polynomials in $x_1, \dots, x_d$:

$$e_k = \sum_{1 \le j_1 < j_2 < \cdots < j_k \le d} x_{j_1} x_{j_2} \dots x_{j_k}$$

To find $\left(R^{\otimes_k d}[Y]/(Y^p - Y - \sum_{i=1}^d f(x_i))\right)^{S_d}$ its enough to express $\sum_{i=1}^d f(x_i)$ in $e_1, \dots, e_d$, this can be done with the newton polynomials, moreover, we claim the following:

Lemma 46. *Let $f(x) = cx^{-m} + a_{-m+1}x^{-m+1} + \cdots a_{-1}x^{-1} + a_0$, define $\alpha(x_1, ..., x_d) = \sum_1^d f(x_i)$, and deonte by $e_1, ..., e_d$ the elementary symmetric polynomials in $x_1, ..., x_d$. If $m < d$ then $\alpha(x_1, ..., x_d) \in k(e_1/e_d, e_2/e_d, ..., e_{d-1}/e_d)$*

Proof. Changing variables $y_i = x_i^{-1}$ for each $i \in \{1, \ldots, d\}$ We get

$$\alpha = \sum_{i=1}^{d} f(x_i) = \sum_{i=1}^{d}\Big(cy_i^m + a_{-m+1}y_i^{m-1} + \cdots + a_{-1}y_i + a_0\Big)$$

Rearranging the sums, we get

$$\alpha = c\sum_{i=1}^{d} y_i^m + a_{-m+1}\sum_{i=1}^{d} y_i^{m-1} + \cdots + a_{-1}\sum_{i=1}^{d} y_i + da_0$$

Let $p_k(y_1, \ldots, y_d) = \sum_{i=1}^d y_i^k$ be the k-th power sum symmetric polynomial. The expression for α is a linear combination of these power sums:

$$\alpha = cp_m(y) + a_{-m+1}p_{m-1}(y) + \cdots + a_{-1}p_1(y) + da_0$$

According to the *Fundamental Theorem of Symmetric Polynomials*, any symmetric polynomial in $y_1, \ldots, y_d$ can be expressed as a polynomial in the elementary symmetric polynomials $e_k(y_1, \ldots, y_d)$. Since $m < d$, α is a polynomial in $e_1(y), e_2(y), \ldots, e_m(y)$. $(y = (y_1, \ldots, y_d))$ The elementary symmetric polynomials in $y_i = 1/x_i$ are related to the elementary symmetric polynomials in x_i as follows:

$$e_k(y_1, \ldots, y_d) = \sum_{1\le i_1<\cdots<i_k\le d} \frac{1}{x_{i_1}\ldots x_{i_k}} = \frac{\sum_{1\le j_1<\cdots<j_{d-k}\le d} x_{j_1}\ldots x_{j_{d-k}}}{x_1x_2\ldots x_d}$$

Thus,

$$e_k(y_1, \ldots, y_d) = \frac{e_{d-k}(x_1, \ldots, x_d)}{e_d(x_1, \ldots, x_d)}$$

which concludes the proof. □

Finally, restricting $\mathcal{P}^{[d]}$ to spec $\hat{K}$ we get by (5) and Theorem 30 the result. (that $\mathcal{P}$ is unramified at the generic point of the exceptional divisor of the blowup).

Kummer Extensions: Few things are different in that case,

Set $R = k[x, x^{-1}]$, and $S = R[X]/(X^n - f)$ where $f = f(x, 1/x) \in R$ In this case we have $char k = p$ and $gcd(p, n) = 1$.

We have

$$\begin{aligned} R^{\otimes_k d} &= k[x_1, x_1^{-1}, \ldots, x_d, x_d^{-1}] \\ S^{\otimes_k d} &= k[x_1, x_1^{-1}, \ldots, x_d, x_d^{-1}][X_1, \ldots, X_d]/(X_1^n - f_1, \ldots, X_d^n - f_d) \\ &= R^{\otimes_k d}[X_1, \ldots, X_d]/(X_1^n - f_1, \ldots, X_d^n - f_d) \end{aligned}$$

Where $f_i = f(x_i, x_i^{-1})$

Next, we want to figure out $\left(S^{\otimes_k d}\right)^{G^{d-1}}$

Recall that the action of $g \in G = \mathbb{Z}/n\mathbb{Z}$ on X is $g(X) = \zeta^g X$ (g correspond to a number $0 \le g \le n-1$). So, the action of $(g_1, \dots, g_{d-1})$ on the generators $(X_1, X_2, \dots, X_{d-1}, X_d)$ Is $X_1 \mapsto \zeta^{g_1} X_1$, $X_i \mapsto \zeta^{g_i - g_{i-1}} X_i$ for $1 < i < d$ and $X_d \mapsto \zeta^{-g_{d-1}} X_d$.

So we see that $Y = X_1 X_2 \dots X_d$ is invariant. And $Y^n - \prod_{i=1}^d f_i$ is irreducible degree n equation for Y, So, like before, we conclude:

$$\left(S^{\otimes_k d}\right)^{G^{d-1}} \cong R^{\otimes_k d}[Y]/(Y^n - \prod_{i=1}^d f_i)$$

The group S_d acts on $R^{\otimes_k d}[Y]/(Y^n - \prod_{i=1}^d f_i)$ by permuting the indices. on the variables x_i, On $Y = \prod_1^d X_i$ it is invaraint. The invaraint subring $(R^{\otimes_k d})^{S_d}$ is simply $k[e_1, e_2, \dots, e_d, e_d^{-1}]$ like before.

The polynomial $F = \prod_{i=1}^d f_i$ is symmetric in $\{x_i\}_1^d$ so it can be expressed as a polynomial $\tilde{F}(e_1, \dots, e_d)$ in the elementary symmetric variables. Hence the qoutient ring is:

$$\left(\frac{k[x_1^{\pm 1}, \dots, t_d^{\pm 1}][Y]}{(Y^n - \prod_{i=1}^d f_i)}\right)^{S_d} \cong \frac{k[e_1, \dots, e_d, e_d^{-1}][Y]}{(Y^n - \tilde{F}(e_1, \dots, e_d))}$$

So we see again, that restricting $\mathcal{P}^{[d]}$ to $\operatorname{spec} \hat{K}$ we get by (5) and Theorem 29, that $\mathcal{P}$ is tamely ramified at the generic point of the exceptional divisor of the blowup. □

So, we conclude

Corollary 47. *In the general settings of a curve C, and $\mathcal{P} \to U = C \setminus \mathfrak{m}$ a G-torsor with ramification bounded by $\mathfrak{n} = nP \subset \mathfrak{m}$ at P, we have:*

1. *If $G = \mathbb{Z}/p\mathbb{Z}$ then $\mathcal{P}^{[\deg \mathfrak{n}]}$ is unramified at $\eta_{\mathfrak{n}}$*
2. *If $\mathcal{P}$ is tamely ramified at $\mathfrak{n}$ then $\mathcal{P}^{[\deg \mathfrak{n}]}$ is tamely ramified at $\eta_{\mathfrak{n}}$*

Next, we generalize by induction to the case of $G = \mathbb{Z}/p^r\mathbb{Z}$.

Theorem 48. *Assume $G = \mathbb{Z}/p^r\mathbb{Z}$. Then $\mathcal{P}^{[\deg(\mathfrak{n})]}$ is unramified at $\eta_{\mathfrak{n}}$*

Proof. Again, we can assume k is algebraically closed. $\mathcal{P}^{[\deg \mathfrak{n}]}$ has bounded ramification at nP by n. Let $R = \widehat{\mathcal{O}_{C,P}}$ and let $K = Frac(R)$ Then $\mathcal{P}^{[\deg \mathfrak{n}]}|_{\operatorname{Spec} K}$ correspond to $\rho : \operatorname{Gal}(K^{sep}/K) \to G$. Let $H = \rho(G^0)$ be the image of the inertia subgroup. Decompose $\mathcal{P} \to U$ to $\mathcal{P} \to \mathcal{P}_{G/H} \to U$ as in Proposition 14, where $\mathcal{P} \to \mathcal{P}_{G/H}$ is an H-torsor and $\mathcal{P}_{G/H} \to U$ is a G/H torsor. Then $\mathcal{P}_{G/H}$ is unramified at P. Thus $\mathcal{P}_{G/H}$ extends to $U' = U \cup \{P\}$, and taking any point $Q \in \mathcal{P}_{G/H}$ over P, results in isomorphism $\widehat{\mathcal{O}_{\mathcal{P}_{G/H},Q}} \cong \widehat{\mathcal{O}_{C,P}}$. But because k is algebraically closed, we even have equality, and deduce that locally the G/H action on $\widehat{\mathcal{O}_{C,P}}$ is trivial. Thus, we can assume $G = H$ and that $\mathcal{P} \to U$ is totally ramified at P.

We proceed by induction on r. The base case $r = 1$ is done by the previous theorem. Decompose $\mathcal{P} \to U$ as $\mathcal{P} \to \mathcal{P}' \to U$ where $\mathcal{P} \to \mathcal{P}'$ is a $\mathbb{Z}/p\mathbb{Z}$-torsor and $\mathcal{P}' \to U$ is a $\mathbb{Z}/p^{r-1}\mathbb{Z}$-torsor.

(1) By the induction hypothesis, $\mathcal{P}'^{[\deg \mathfrak{n}]}$ is unramified at $\eta_{\mathfrak{n}}$.

Let C' be the normal proper model of $\mathcal{P}'$ viewed as a scheme, and let $\pi : C' \to C$ be the extension of $\mathcal{P}' \to U$. Let $\mathfrak{n}' = \pi^{-1}(\mathfrak{n})$ be the pullback of $\mathfrak{n}$ to C'. Let $\tilde{\mathfrak{n}}$ be the pullback of $\mathfrak{n}$ to C'.

Then, following Lemma 44, and the discussion surrounding it, with U' being $\mathcal{P}'$, we get that

(2) $\mathcal{P}'^{(\deg \mathfrak{n})} \to \mathcal{P}'^{[\deg \mathfrak{n}]} \hookrightarrow C'^{[\deg \mathfrak{n}]}$ is unramified at codimension one points.

And since $\mathcal{P} \to \mathcal{P}'$ is a $\mathbb{Z}/p\mathbb{Z}$-torsor, we can apply the previous theorem to deduce that $\mathcal{P}^{[\deg \mathfrak{n}]} \to \mathcal{P}'^{(\deg \mathfrak{n})}$ is unramified at $\eta_{\mathfrak{n}}$.

Not clear how to finish here without analyzing the ramification even further. Thus, take a subgroup $H \subset G$ such that $G/H \cong \mathbb{Z}/p\mathbb{Z}$ and decompose $\mathcal{P} \to U$ as: $\square$

Finally, **Proof of Theorem theorem 37**: Idea: decompose G be the structure theorem, and finish by the above.

3.3 Extending To Product of Blowups

We conclude this sesction by proving sume auxilary results that would be useful in Section 4.

Let X and Y be smooth schemes over a field k, and let $x \in X$ and $y \in Y$ be closed points. We denote the blowups of these schemes at the given points by $\pi_X : \mathrm{Bl}_x(X) \to X$ and $\pi_Y : \mathrm{Bl}_y(Y) \to Y$. Furthermore, let $\pi_{X\times Y} : \mathrm{Bl}_{(x,y)}(X \times_k Y) \to X \times_k Y$ be the blowup of the product scheme at the point (x, y). We denote by E_X, E_Y, and $E_{X\times Y}$ the respective exceptional divisors, and let η_X, η_Y, and $\eta_{X\times Y}$ be their generic points.

In this section, we establish the following result concerning the stability of ramification bounds under the external product of torsors.

Proposition 49. *Let G be a finite abelian group. Suppose $\mathcal{G}_X$ and $\mathcal{G}_Y$ are G-torsors defined on open subsets $U_X \subset \mathrm{Bl}_x(X)$ and $U_Y \subset \mathrm{Bl}_y(Y)$ that are disjoint from the exceptional divisors. If the ramification of $\mathcal{G}_X$ at η_X and $\mathcal{G}_Y$ at η_Y is bounded by r, then the external product torsors*

$$\mathcal{G}_{X\times Y} := pr_1^{-1}\mathcal{G}_X \otimes pr_2^{-1}\mathcal{G}_Y$$

has ramification bounded by r at the generic point $\eta_{X\times Y}$ of the exceptional divisor in the product blowup.

The proposition is purely local in nature, it suffices to consider the case where X and Y are affine. More precisely, by the smoothness of X and Y, we may restrict our attention to open neighborhoods of x and y that are etale over affine spaces. The rest of this section treats that case.

The Affine Case

Let $X = \mathbb{A}^n_k$ be the affine n-space over a field k, and let $0 \in X$ be the origin. Let $\tilde{X} = \mathrm{Bl}_0(X)$ be the blowup of X at the origin. Recall that $\tilde{X} \subset X \times_k \mathbb{P}^{n-1}_k$ is defined by the equations $x_i u_j = x_j u_i$, where $[u_1 : \cdots : u_n]$ are the homogeneous coordinates of $\mathbb{P}^{n-1}_k$. The exceptional divisor $E \subset \tilde{X}$ is the fiber over the origin, $E = \{(0, [u_1 : \cdots : u_n])\}$, which is of codimension 1 in $\tilde{X}$. Let $\eta \in E$ be the generic point of E, and let $R = \mathcal{O}_{\tilde{X},\eta}$ be the associated local ring. This ring R is a discrete valuation ring (DVR) with fraction field $K = K(X) = k(x_1, \dots, x_n)$.

On the affine chart U_1 where $u_1 \neq 0$, we have $x_i = \frac{u_i}{u_1}x_1$. The coordinate ring is:

$$\mathcal{O}_{\tilde{X}}(U_1) = k\left[x_1, \frac{u_2}{u_1}, \dots, \frac{u_n}{u_1}\right]$$

In this chart, the generic point η corresponds to the prime ideal $\mathfrak{p}_1 = (x_1)$. Thus, the local ring is $R = k[x_1, \frac{u_2}{u_1}, \dots, \frac{u_n}{u_1}]_{(x_1)}$. The residue field is $\kappa(\eta) = k(\frac{u_2}{u_1}, \dots, \frac{u_n}{u_1})$, and the completion of R with respect to its maximal ideal is:

$$\hat{R} = \kappa(\eta)[[x_1]]$$

In this local ring, x_1 is a uniformizer. Note that any x_i (for $i > 1$) can also serve as a uniformizer, as $x_i = (\frac{u_i}{u_1})x_1$ and $\frac{u_i}{u_1}$ is a unit in R.

For any monomial $M = x_1^{a_1} \dots x_n^{a_n} \in K$, we can write:

$$M = x_1^{\sum a_i} \left(\frac{u_2}{u_1}\right)^{a_2} \dots \left(\frac{u_n}{u_1}\right)^{a_n}$$

Since the term in parentheses is a unit in R, the valuation ν_E associated with E satisfies:

$$\nu_E(M) = \sum a_i = \deg M$$

Consequently, for any polynomial $f = f_d + f_{d+1} + \dots + f_l$, where f_i is the homogeneous part of degree i, we have $\nu_E(f) = d$ (the order of vanishing at the origin).

The Product Case

Now, let $X = \mathbb{A}^n$ and $Y = \mathbb{A}^m$ with origins $x = 0$ and $y = 0$. As before, $\mathrm{Bl}_0(X) \subset X \times \mathbb{P}^{n-1}$ and $\mathrm{Bl}_0(Y) \subset Y \times \mathbb{P}^{m-1}$ have exceptional divisors E_X and E_Y respectively. Consider the product $X \times_k Y \cong \mathbb{A}_k^{n+m}$. The blowup of the product at the origin $(0,0)$, denoted $\mathrm{Bl}_{(0,0)}(X \times_k Y)$, is a subscheme of $(X \times Y) \times \mathbb{P}^{n+m-1}$ defined by:

$$\begin{cases} x_i w_j = x_j w_i & 1 \le i, j \le n \\ y_k w_{n+l} = y_l w_{n+k} & 1 \le k, l \le m \\ x_i w_{n+j} = y_j w_i & 1 \le i \le n,\ 1 \le j \le m \end{cases}$$

where $[w_1 : \dots : w_{n+m}]$ are the homogeneous coordinates of $\mathbb{P}^{n+m-1}$. The exceptional divisor $E_{X \times Y}$ is isomorphic to $\mathbb{P}^{n+m-1}$.

Comparison of Blowups

Both $\mathrm{Bl}_0(X) \times_k \mathrm{Bl}_0(Y)$ and $\mathrm{Bl}_{(0,0)}(X \times_k Y)$ are birational to $X \times_k Y$. They share a common dense open set $\tilde{U}$ defined by the condition that neither the X-coordinates nor the Y-coordinates vanish simultaneously in the projective space:

$$\tilde{U} = \{((x, y), [w_1 : \dots : w_{n+m}]) \mid (w_1, \dots, w_n) \neq 0 \text{ and } (w_{n+1}, \dots, w_{n+m}) \neq 0\}$$

This yields a diagram of open immersions:

$$\begin{tikzcd} & \tilde{U} \arrow[dl, "f_1"'] \arrow[dr, hook, "f_2"] & \\ \mathrm{Bl}_0(X) \times_k \mathrm{Bl}_0(Y) & & \mathrm{Bl}_{(0,0)}(X \times_k Y) \end{tikzcd}$$

where f_1 maps the coordinates to the respective projectivizations $[w_1 : \dots : w_n]$ and $[w_{n+1} : \dots : w_{n+m}]$. Also note that the generic point $\eta_{X \times Y}$ of $E_{X \times_k Y}$ is inside $\tilde{U}$

Extensions of DVRs

Let S be the local ring of the generic point $\eta_{X\times Y}$ of $E_{X\times Y}$ in $\tilde{U}$. On the chart where $w_1 \neq 0$ and $w_{n+1} \neq 0$, we have $x_1 = (\frac{w_1}{w_{n+1}})y_1$. Since $\frac{w_1}{w_{n+1}}$ is a unit in this chart, x_1 and y_1 are equivalent as uniformizers. We have:

$$S = k\left[x_1, \frac{w_2}{w_1} \cdots, \frac{w_n}{w_1}, \frac{w_{n+1}}{w_1} \cdots \frac{w_{n+m}}{w_1}\right]_{(x_1)} = k\left[\frac{w_1}{w_{n+1}}, \frac{w_2}{w_{n+1}} \cdots \frac{w_n}{w_{n+1}}, y_1 \cdots \frac{w_{n+m}}{w_{n+1}}\right]_{(y_1)}$$

$$k(\eta_{X\times Y}) = k\left(\frac{w_2}{w_1} \cdots, \frac{w_n}{w_1}, \frac{w_{n+1}}{w_1} \cdots \frac{w_{n+m}}{w_1}\right)$$

Let R_X be the local ring of the exceptional divisor E_X in $\mathrm{Bl}_0(X)$. The pullback of $E_X \times_k \mathrm{Bl}_0(Y)$ along f_1 induces an extension of DVRs $R_X \hookrightarrow S$. Which is:

1. Weakly Unramified: x_1 is a uniformizer in both R_X and S, so the ramification index is $e = 1$.
2. Residually Transcendental: The residue field extension $\kappa(\eta_X) \subset \kappa(\eta)$ is:

   $$k\left(\frac{w_2}{w_1}, \dots, \frac{w_n}{w_1}\right) \subset k\left(\frac{w_2}{w_1}, \dots, \frac{w_n}{w_1}, \frac{w_{n+1}}{w_1}, \dots, \frac{w_{n+m}}{w_1}\right)$$

   Hence separable.

Since this extension is generated by transcendental elements, it is separable and formally smooth at the maximal ideal ([Stacks, Tag 09E7]).

Ramification of G-Torsors

Let G be a finite abelian group. Let $\mathcal{Q}$ be a G-torsor on an open $U \subset \mathrm{Bl}_0(X)$ disjoint from E_X. Let $V \subset \mathrm{Bl}_0(Y)$ be an open subscheme, and let $\pi_X : \mathrm{Bl}_0(X) \times_k V \to \mathrm{Bl}_0(X)$ be the projection onto the first factor. By restricting this projection to $U \times_k V$, we obtain the pullback G-torsor:

$$\pi_X^{-1}(\mathcal{Q}) \cong \mathcal{Q} \times_k V$$

which is defined on the open subset $U \times_k V \subset \mathrm{Bl}_0(X) \times_k \mathrm{Bl}_0(Y)$. The extension of local rings $R_X \to S$ is weakly unramified (the ramification index $e = 1$) and residually transcendental with seprable residue field exntesion. Under these conditions the ramification filtration is preserved. Therefore, the pullback $\pi_X^{-1}(\mathcal{Q})$ has ramification bounded by r at the generic point $\eta_{X\times Y}$ of the exceptional divisor $E_{X\times Y}$ if and only if the original torsor $\mathcal{Q}$ has ramification bounded by r at the generic point η_X of E_X

And we finish by Lemma 42.

4 Geometric Class Field Theory

The main theorem of this section is

Theorem 50. *Let Λ be a finite ring of cardinality invertible in k, and let $\mathcal{F}$ be an étale sheaf of Λ-modules, locally free of rank 1 on U, with ramification bounded by $\mathfrak{m}$. Considering $U^{(d)}$ as an open subscheme of the blowup $\tilde{C}_{\mathfrak{m}}^{(d)}$ of $C^{(d)}$, we have that for sufficiently large integer d, $\mathcal{F}^{[d]}$ is tamely ramified on $H = \tilde{C}_{\mathfrak{m}}^{(d)} \setminus U^{(d)} = E_0 \times_{C^{(d)}} \tilde{C}_{\mathfrak{m}}^{(d)}$.*

From which we will deduce Theorem 2.

We start by recalling the necessarily material, and then move to proving this theorem.

4.1 Etale Fundamental Groups and Tame Fundamental Groups

We recall the definition and basic properties of the etale fundamental group, following stacks project [Stacks, Tag 0BQ6]

Proposition 51 ([Stacks, Tag 0C0J])**.** *Let $f : X \to S$ be a flat proper morphism of finite presentation whose geometric fibres are connected and reduced. Assume S is connected and let $\overline{s}$ be a geometric point of S. Then there is an exact sequence*

$$\pi_1(X_{\overline{s}}) \to \pi_1(X) \to \pi_1(S) \to 1$$

of fundamental groups.

Corollary 52. *Let $f : X \to S$ be a proper smooth morphism of finite presentation whose geometric fibres are connected. Assume S is connected and let $\overline{s}$ be a geometric point of S. Then there is an exact sequence*

$$\pi_1(X_{\overline{s}}) \to \pi_1(X) \to \pi_1(S) \to 1$$

of fundamental groups.

4.2 Generalized Picard Scheme

We recall the notion of generalized Jacobian varieties and study their fundamental properties. The material presented here is primarily adapted from [Gui19] and [Tak19]. For further background on the general theory of abelian varieties and Jacobians, the reader may also consult [Mil08]. Let S be a scheme and let C be a projective smooth S-scheme whose geometric fibers are connected and of dimension 1. Let $\mathfrak{m}$ be a modulus on C, defined as an effective Cartier divisor of C/S (i.e., a closed subscheme of C which is finite flat of finite presentation over S). We denote the projection $C \times_S T \to T$ by pr for any S-scheme T.

The Functor of Points

Let d be an integer. For an S-scheme T, we consider the set of data $(\mathcal{L}, \psi)$ where:

- $\mathcal{L}$ is an invertible sheaf of degree d on C_T.
- $\psi : \mathcal{O}_{\mathfrak{m}_T} \xrightarrow{\sim} \mathcal{L}|_{\mathfrak{m}_T}$ is a trivialization of $\mathcal{L}$ along the modulus.

Two such pairs $(\mathcal{L}, \psi)$ and $(\mathcal{L}', \psi')$ are said to be isomorphic if there exists an isomorphism of invertible sheaves $f : \mathcal{L} \to \mathcal{L}'$ such that the following diagram commutes:

$$\begin{tikzcd} & \mathcal{O}_{\mathfrak{m}_T} \arrow[dl, "\psi'"'] \arrow[dr, "\psi"] & \\ \mathcal{L}'|_{\mathfrak{m}_T} \arrow[rr, "f|_{\mathfrak{m}_T}"] & & \mathcal{L}|_{\mathfrak{m}_T} \end{tikzcd}$$

We define the presheaf $\mathrm{Pic}^{d,\mathrm{pre}}_{C,\mathfrak{m}}$ on Sch/S by assigning to T the set of isomorphism classes of such pairs. Let $\mathrm{Pic}^d_{C,\mathfrak{m}}$ denote the étale sheafification of this presheaf.

Representability and Structure

The fundamental properties of this functor are as follows:

1. $\mathrm{Pic}^d_{C,\mathfrak{m}}$ is represented by an S-scheme. (Note: If $\mathfrak{m}$ is faithfully flat over S, the presheaf is already a étale sheaf).
2. $\mathrm{Pic}^0_{C,\mathfrak{m}}$ is a smooth commutative group S-scheme with geometrically connected fibers, referred to as the *generalized Jacobian variety* of C with modulus $\mathfrak{m}$.
3. For any d, $\mathrm{Pic}^d_{C,\mathfrak{m}}$ is a $\mathrm{Pic}^0_{C,\mathfrak{m}}$-torsor.

In the case where $\mathfrak{m} = 0$, we recover the standard Jacobian variety, denoted simply as Pic^d_C.

Relation to the Standard Jacobian

We now examine the behavior of the generalized Picard scheme under the variation of the modulus. By viewing the structure along the modulus as an additional rigidification, we obtain natural transition maps corresponding to the inclusion of moduli.

Let $\mathfrak{m}_1$ and $\mathfrak{m}_2$ be moduli such that $\mathfrak{m}_1 \subset \mathfrak{m}_2$. There exists a natural map

$$\mathrm{Pic}^d_{C,\mathfrak{m}_2} \to \mathrm{Pic}^d_{C,\mathfrak{m}_1}$$

obtained by restricting the isomorphism ψ. Since $\mathfrak{m}_2$ is a finite S-scheme, this map is a surjection as a morphism of étale sheaves. In particular, for any modulus $\mathfrak{m}$, there is a natural surjective morphism of étale sheaves:

$$\mathrm{Pic}^d_{C,\mathfrak{m}} \to \mathrm{Pic}^d_C.$$

Local Freeness and Base Change

Let $\mathfrak{m}$ be a modulus which is everywhere strictly positive. Let g denote the genus of C, which is a locally constant function on S. We restrict our attention to degrees d satisfying the condition:

$$d \geq \max\{2g - 1 + \deg \mathfrak{m}, \deg \mathfrak{m}\}. \tag{6}$$

Assuming S is quasi-compact, such a d always exists.

Fix an integer d satisfying the condition above. Let T be an S-scheme and let $\mathcal{L}$ be an invertible sheaf of degree d on C_T. One can show that the pushforwards $\mathrm{pr}_*\mathcal{L}$ and $\mathrm{pr}_*\mathcal{L}(-\mathfrak{m})$ are locally free sheaves and their formations commute with any base change. Explicitly, for any morphism of S-schemes $f : T' \to T$, the base change morphisms are isomorphisms:

$$f^*\mathrm{pr}_*\mathcal{L} \xrightarrow{\sim} \mathrm{pr}_* f^*\mathcal{L}$$

and

$$f^*\mathrm{pr}_*(\mathcal{L}(-\mathfrak{m})) \xrightarrow{\sim} \mathrm{pr}_* f^*(\mathcal{L}(-\mathfrak{m})).$$

In particular, following [Gui19], if $\mathcal{L}$ is invertible $\mathcal{O}_C$-module with degree d satisfying 6 on each fiber of f then, $\mathrm{pr}_*\mathcal{L}$ is a locally free $\mathcal{O}_S$-module of rank $d - g + 1$.

For further background and verification of these constructions, we refer the reader to Milne's notes on abelian Varieties ([Mil08]).

4.3 The Abel-Jacobi Morphism and its Fibers

Let $U = C \backslash \mathfrak{m}$ be the complement of the modulus in C. The effective cartier divisors of degree d which are prime to $\mathfrak{m}$ are parameterized by the symmetric power $\mathrm{Sym}^d_S(U) = U^{(d)}$ over S (See [Gui19] Proposition 4.12, [Mil08] Theorem 3.13). For any such divisor $D \in U^{(d)}$, the associated line bundle $\mathcal{O}_C(D)$ admits a canonical trivialization along $\mathfrak{m}$. Specifically, the canonical section 1_D is regular and non-vanishing on $\mathfrak{m}$ because $\mathrm{supp}(D) \cap \mathrm{supp}(\mathfrak{m}) = \emptyset$. This section restricts to a nowhere-vanishing section on the subscheme $\mathfrak{m}$, thereby determining a trivialization $\psi_D^{-1} : \mathcal{O}_C(D)|_{\mathfrak{m}} \xrightarrow{\sim} \mathcal{O}_{\mathfrak{m}}$. This is done functorially in families, yielding a morphism from the symmetric power to the generalized Picard scheme (over S):

$$\Phi_d : U^{(d)} \to \mathrm{Pic}^d_{C,\mathfrak{m}}, \quad D \mapsto [(\mathcal{O}_C(D), \psi_D)], \tag{7}$$

When $\mathfrak{m} = 0$, $d \geq max\{2g-1, 0\}$ and C admits a section over S, $C^{(d)}$ is a projective space bundle over Pic^d_C, It is proper, surjective with geometrically connected fibers.

Guignard ([Gui19] Theorem 4.14) proves that for $\mathfrak{m} > 0$ and d satisfying (6), the Abel-Jacobi morphism Φ_d is surjective smooth of relative dimension $d - \deg \mathfrak{m} - g + 1$, with geometrically connected fibers.

When $S = \mathrm{Spec}(k)$, the geometric-fibers of Φ_d are well understood:

Theorem 53. *Assuming $S = \mathrm{Spec}(k)$ and $d \geq max\{2g - 1 + \deg \mathfrak{m}, \deg \mathfrak{m}\}$. Then, the geometric-fibers of the Abel-Jacobi morphism*

$$\Phi_d : U^{(d)} \to \mathrm{Pic}^d_{C,\mathfrak{m}}$$

over any point are isomorphic to

$$\begin{cases} \mathbb{A}^{d-\deg \mathfrak{m} - g+1}_{k^{sep}} & \textit{if } m > 0 \\ \mathbb{P}^{d-g}_{k^{sep}} & \textit{if } m = 0 \end{cases}$$

In both cases Φ_d is a fibration in affine spaces or projective spaces, depending on whether $\mathfrak{m}$ is non-zero or zero.

Proof. see [Ten15] Propositions 3.13-3.14, or [Tót11] Prop 2.1.4:

□

4.4 Compactification of Blowup of Symmetric Powers of a Curve

We recall that our objective is to descend the local system $\mathcal{F}^{[d]}$ from $U^{(d)}$ to $\mathrm{Pic}^d_{C,\mathfrak{m}}$ along the Abel-Jacobi map Φ_d:

$$\begin{tikzcd} \mathcal{F}^{[d]} \arrow[d] & \\ U^{(d)} \arrow[r, "\Phi_d"] & \mathrm{Pic}^d_{C,\mathfrak{m}} \end{tikzcd}$$

(Here, the purple arrow emphasizes that the morphism is of sheaves on the étale site).

However, we encounter an obstruction: in the case we are considering ($\mathfrak{m} > 0$), the fibers of Φ_d are affine spaces (of the same degree) rather than the better-behaved projective spaces. This hint that a solution to this problem is to compactify the morphism to yield projective fibers.

This section describes the result of the compactification constructed by [Tak19] via the method of blowup.

Let $\mathfrak{m} = \sum_{i=1}^{n} k_P P$ with $\deg P = d_P$ be a modulus on C, and let d satisfy (6). Takeuchi ([Tak19]) defines $Z_0 = Z_0(\mathfrak{m}, d)$ as the closed subscheme of $C^{(d)}$ defined by the map $C^{(d-\deg \mathfrak{m})} \to C^{(d)}$ adding $\mathfrak{m}$. He also defines $X_{\mathfrak{m},d}$ as the blowup of $C^{(d)}$ along Z_0. Let $E_0 = E_{\mathfrak{m},d} = Z_0(\mathfrak{m}, d) \times_{C^{(d)}} X_{\mathfrak{m},d}$ be the exceptional divisor of the blowup. It is irreducible of codimension 1, and we let $\eta_0 = \eta_{\mathfrak{m},d}$ be its generic point.

Diagrammatically:

$$\begin{tikzcd} \overline{\{\eta_0\}} = E_0 \arrow[r] \arrow[d] & X_{\mathfrak{m},d} \arrow[d, "\pi"] \\ Z_0 \arrow[r, "c.i"] & C^{(d)} \end{tikzcd}$$

Incoporating $U^{(d)}$, the local system $\mathcal{F}^{[d]}$ and the Abel-Jacobi map, we have:

$$\begin{tikzcd} & & \mathcal{F}^{[d]} \arrow[d] & \\ \overline{\{\eta_0\}} = E_0 \arrow[r] \arrow[d] & X_{\mathfrak{m},d} \arrow[d, "\pi"] & U^{(d)} \arrow[l, hook'] \arrow[r, "\Phi_d"] \arrow[dl, hook'] & \mathrm{Pic}^d_{C,\mathfrak{m}} \\ Z_0 \arrow[r, "c.i"] & C^{(d)} & & \end{tikzcd}$$

In Section 3 of [Tak19] Takeuchi constructs, for d satisfying (6) a compactification denoted by $\tilde{C}^{(d)}_{\mathfrak{m}}$ and proves the following:

Theorem 54 (Takeuchi)**.** *The scheme $\tilde{C}^{(d)}_{\mathfrak{m}}$ is an open subscheme of $X_{\mathfrak{m},d}$ containing $U^{(d)}$. The morphism $\Phi_d : U^{(d)} \to \mathrm{Pic}^d_{C,\mathfrak{m}}$ extends to a morphism $\tilde{\Phi}_d : \tilde{C}^{(d)}_{\mathfrak{m}} \to \mathrm{Pic}^d_{C,\mathfrak{m}}$ which makes $\tilde{C}^{(d)}_{\mathfrak{m}}$ a projective space bundle over $\mathrm{Pic}^d_{C,\mathfrak{m}}$. Furthermore, the complement of $U^{(d)}$ in $\tilde{C}^{(d)}_{\mathfrak{m}}$ is isomorphic to the fiber product $E_0 \times_{C^{(d)}} \tilde{C}^{(d)}_{\mathfrak{m}}$.*

Diagrammatically we have:

$$\begin{tikzcd}
 & & \mathcal{F}^{[d]} \arrow[d] \\
E_0 \times_{C^{(d)}} \tilde{C}_{\mathfrak{m}}^{(d)} \arrow[r] \arrow[d] & \tilde{C}_{\mathfrak{m}}^{(d)} \arrow[d, hook] \arrow[dr, "\tilde{\Phi}_d"] & U^{(d)} \arrow[l, hook'] \arrow[d, "\Phi_d"] \\
\overline{\{\eta_0\}} = E_0 \arrow[r] \arrow[d] & X_{\mathfrak{m},d} \arrow[d, "\pi"] & \mathrm{Pic}^d_{C,\mathfrak{m}} \\
Z_0 \arrow[r, "c.i"] & C^{(d)} &
\end{tikzcd}$$

Also note that $E_0 \times_{C^{(d)}} \tilde{C}_{\mathfrak{m}}^{(d)} = Z_0 \times_{C^{(d)}} \tilde{C}_{\mathfrak{m}}^{(d)}$

4.5 Proof of Theorem 50

In what follows we reduce Theorem 50 to Theorem 37

4.5.1 Reduction Lemmas

The following lemma is adapted from [Tak19] (Lemma 4.1)

Lemma 55. *Let C be a projective, smooth, and geometrically connected curve over a perfect field k. Let $\mathfrak{m} = \sum_{i=1}^{r} k_i P_i$ be an effective divisor where $P_1, \ldots, P_r$ are distinct closed points. Let $U = C \setminus \mathfrak{m}$ and let $d \geq \deg \mathfrak{m}$.*

Suppose $\mathfrak{n}_1, \ldots, \mathfrak{n}_l$ are pairwise coprime submoduli of $\mathfrak{m}$ such that $\mathfrak{m} = \sum_{j=1}^{l} \mathfrak{n}_j$. Consider the summation morphism:

$$\pi : C^{(\deg \mathfrak{n}_1)} \times_k \cdots \times_k C^{(\deg \mathfrak{n}_l)} \times_k C^{(d-\deg \mathfrak{m})} \longrightarrow C^{(d)}$$

defined by $(D_1, \ldots, D_l, D_{extra}) \mapsto \sum_{j=1}^{l} D_j + D_{extra}$.

Then π is étale at the generic point of the closed subvariety

$$V = \{\mathfrak{n}_1\} \times_k \cdots \times_k \{\mathfrak{n}_l\} \times_k C^{(d-\deg \mathfrak{m})}$$

inside the domain $C^{(\deg \mathfrak{n}_1)} \times_k \cdots \times_k C^{(\deg \mathfrak{n}_l)} \times_k C^{(d-\deg \mathfrak{m})}$.

Proof. We may assume that k is algebraically closed (hence $\deg P_i = 1$ for all i). By miracle flatness π is flat, it is quasi-finite and projective as a map between projecive spaces. so we condlude π is finite and flat. It is enough to show that there exists a closed point Q of $\mathfrak{n}_1 + \ldots \mathfrak{n}_l + C^{(d-\deg \mathfrak{m})} \subset C^{(d)}$ over which there are $\deg \pi$ points on $C^{(\mathfrak{n}_1)} \times_k \cdots \times_k C^{(\mathfrak{n}_l)} \times_k C^{(d-\deg \mathfrak{m})}$. (Because it will be unramified at this point and thus also at the generic point of V.) Choose Q as a point corresponding to a divisor $\mathfrak{n}_1 + \cdots + \mathfrak{n}_l + P_{r+1} + \cdots + P_{r+d-\deg \mathfrak{m}}$, where $P_1, \ldots, P_{r+d-\deg \mathfrak{m}}$ are distinct points of $U(k)$. $\square$

Corollary 56. *The morphism $C^{(\deg \mathfrak{m})} \times_k C^{(d-\deg \mathfrak{m})} \xrightarrow{\pi} C^{(d)}$ is finite flat everywhere, and étale at the generic point of the closed subvariety $Z_{\mathfrak{m}} \times_k C^{(d-\deg \mathfrak{m})} \subset C^{(\deg \mathfrak{m})} \times_k C^{(d-\deg \mathfrak{m})}$.*

Following from this, we look at the following diagram, coming from the flat base change $C^{(d-\deg\mathfrak{m})} \to \mathrm{Spec}(k)$ (Proposition 19) of (4):

$$\begin{tikzcd} E_{\mathfrak{m}} \times_k C^{(d-\deg\mathfrak{m})} \arrow[r] \arrow[d] & X_{\mathfrak{m}} \times_k C^{(d-\deg\mathfrak{m})} \arrow[d] & \mathcal{F}^{[\deg\mathfrak{m}]} \times_k C^{(d-\deg\mathfrak{m})} \arrow[d] \\ Z_{\mathfrak{m}} \times_k C^{(d-\deg\mathfrak{m})} \arrow[r] & C^{(\deg\mathfrak{m})} \times_k C^{(d-\deg\mathfrak{m})} & U^{(\deg\mathfrak{m})} \times_k C^{(d-\deg\mathfrak{m})} \arrow[ul] \end{tikzcd}$$

Note that $U^{(\deg\mathfrak{m})} \times_k C^{(d-\deg\mathfrak{m})}$ is dense open subscheme of $X_{\mathfrak{m}} \times_k C^{(d-\deg\mathfrak{m})}$, And $E_{\mathfrak{m}} \times_k C^{(d-\deg\mathfrak{m})}$ is a prime divisor of $X_{\mathfrak{m}} \times_k C^{(d-\deg\mathfrak{m})}$. Hence it is well defined question according to Definition 38 to ask whether $\mathcal{F}^{[\deg\mathfrak{m}]} \times_k C^{(d-\deg\mathfrak{m})}$ is tamely ramified at the generic point θ of $E_{\mathfrak{m}} \times_k C^{(d-\deg\mathfrak{m})}$.

Lemma 57. *If $\mathcal{F}^{[\deg\mathfrak{m}]}$ is tamely ramified at $\eta_{\mathfrak{m}}$, then $\mathcal{F}^{[\deg\mathfrak{m}]} \times_k C^{(d-\deg\mathfrak{m})}$ is tamely ramified at the generic point θ of $E_{\mathfrak{m}} \times_k C^{(d-\deg\mathfrak{m})}$.*

Proof. This follows from [Stacks, Tag 0EYD] □

Replacing $C^{(d-\deg\mathfrak{m})}$ with the dense open subscheme $U^{(d-\deg\mathfrak{m})} \subset C^{(d-\deg\mathfrak{m})}$, we get that the G-torsor $p_1^{-1}\mathcal{P}^{[\deg\mathfrak{m}]}$ ($\mathcal{P}$ correspodns to $\mathcal{F}$ under Proposition 8) is tamely ramified at θ the generic point of $E_{\mathfrak{m}} \times_k U^{(d-\deg\mathfrak{m})} \subset U^{(\deg\mathfrak{m})} \times_k U^{(d-\deg\mathfrak{m})}$, where $p_1 : U^{(\deg\mathfrak{m})} \times_k U^{(d-\deg\mathfrak{m})} \to U^{(\deg\mathfrak{m})}$ is the projection to the first factor.

Looking at the second projection $p_2 : X_{\mathfrak{m}} \times_k U^{(d-\deg\mathfrak{m})} \to U^{(d-\deg\mathfrak{m})}$, and the fact that $\mathcal{P}^{[d-\deg\mathfrak{m}]}$ is étale on $U^{(d-\deg\mathfrak{m})}$ we get that $p_2^{-1}\mathcal{P}^{[d-\deg\mathfrak{m}]} = X_{\mathfrak{m}} \times_k \mathcal{P}^{[d-\deg\mathfrak{m}]}$ is étale on $X_{\mathfrak{m}} \times_k U^{(d-\deg\mathfrak{m})}$. Hence, its restriction to $U^{(\deg\mathfrak{m})} \times_k U^{(d-\deg\mathfrak{m})}$ is unramified at θ.

Thus, by the following lemma, we conclude that $\mathcal{P}^{[\deg\mathfrak{m}]} \boxtimes \mathcal{P}^{[d-\deg\mathfrak{m}]} = p_1^{-1}\mathcal{P}^{[\deg\mathfrak{m}]} \otimes^G p_2^{-1}\mathcal{P}^{[d-\deg\mathfrak{m}]}$ is tamely ramified at θ the generic point of $E_{\mathfrak{m}} \times_k U^{(d-\deg\mathfrak{m})}$.

Lemma 58. *Let $X \to \mathrm{Spec}\, k$ be a scheme over a field k, and let $\mathcal{P}_1, \mathcal{P}_2$ be two G-torsors on U_{et}. Let ξ be the generic point of a prime divisor $D \subset X$. If $\mathcal{P}_1$ is tamely ramified at ξ, and $\mathcal{P}_2$ is unramified at ξ, then the product $\mathcal{P}_1 \otimes^G \mathcal{P}_2$ is tamely ramified at ξ.*

Proof. This follows from Lemma 42. □

Combining this with Corollary 56 we get

Corollary 59. *Let $\mathfrak{m}$ be a modulus as above, and let $\eta_{\mathfrak{m}}$ be the generic point of $E_{\mathfrak{m}}$. Let $\mathcal{P}$ be a G-torsor on U_{et} with ramification bounded by $\mathfrak{m}$. Assume $\mathcal{P}^{[\deg\mathfrak{m}]}$ is tamely ramified at $\eta_{\mathfrak{m}}$. Then $\mathcal{P}^{[\deg\mathfrak{m}]} \boxtimes \mathcal{P}^{[d-\deg\mathfrak{m}]}$ is tamely ramified at the generic point θ of $E_{\mathfrak{m}} \times_k C^{(d-\deg\mathfrak{m})} \subset C^{(\deg\mathfrak{m})} \times_k C^{(d-\deg\mathfrak{m})}$, and $\mathcal{P}^{[d]}$ is tamely ramified at the generic point $\eta_0 = \eta_{\mathfrak{m},d}$ of $E_0 = E_{\mathfrak{m},d}$*

Proof. The first assertion, that $\mathcal{P}^{[\deg\mathfrak{m}]} \boxtimes \mathcal{P}^{[d-\deg\mathfrak{m}]}$ is tamely ramified at θ, follows from the preceding discussion. Thus, it remains to show that $\mathcal{P}^{[d]}$ is tamely ramified at η_0.

Consider the blowup diagram defining $X_{\mathfrak{m},d}$:

$$\begin{tikzcd} \overline{\{\eta_0\}} = E_0 \arrow[r] \arrow[d] & X_{\mathfrak{m},d} \arrow[d, "\pi"] \\ Z_0 \arrow[r, "c.i"] & C^{(d)} \end{tikzcd}$$

By performing a base change along the flat addition map $+ : C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})} \to C^{(d)}$, we obtain the following commutative diagram:

$$\begin{tikzcd}
& \left(C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}\right) \times_{C^{(d)}} E_0 \arrow[r] \arrow[d] & \overline{\{\eta_0\}} = E_0 \arrow[d] \\
& \left(C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}\right) \times_{C^{(d)}} X_{\mathfrak{m},d} \arrow[r] \arrow[d] & X_{\mathfrak{m},d} \arrow[d, "\pi"] \\
\mathfrak{m} \times_k U^{(d-\deg \mathfrak{m})} \arrow[r, "c.i"] & C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})} \arrow[r, "+"] & C^{(d)}
\end{tikzcd}$$

Since blowups commute with flat base change, and the inverse image of the center Z_0 under the map $+$ is $\mathfrak{m} \times_k U^{(d-\deg \mathfrak{m})}$, the scheme $\left(C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}\right) \times_{C^{(d)}} X_{\mathfrak{m},d}$ is the blowup of $C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}$ along $\mathfrak{m} \times_k U^{(d-\deg \mathfrak{m})}$. This, in turn, is isomorphic to the base change of the blowup $X_{\mathfrak{m}}$ (of $C^{(\deg \mathfrak{m})}$ along $\mathfrak{m}$) via the (flat) projection $C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})} \to C^{(\deg \mathfrak{m})}$.

Assembling these facts, we obtain the following Cartesian square:

$$\begin{tikzcd}
& E_{\mathfrak{m}} \times U^{(d-\deg \mathfrak{m})} \arrow[r, "\tilde{+}"] \arrow[d] & \overline{\{\eta_0\}} = E_0 \arrow[d] \\
& X_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})} \arrow[r, "\tilde{+}"] \arrow[d] & X_{\mathfrak{m},d} \arrow[d, "\pi"] \\
\mathfrak{m} \times_k U^{(d-\deg \mathfrak{m})} \arrow[r, "c.i"] & C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})} \arrow[r, "+"] & C^{(d)}
\end{tikzcd}$$

The point θ defined in the Corollary is the generic point of $E_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})}$. Let η be the generic point of $\mathfrak{m} \times_k U^{(d-\deg \mathfrak{m})}$. By Corollary 56, the map $+$ is étale at η. Consequently, the lifted map $\tilde{+}$ is étale at θ. Given the isomorphism $(+^{-1})(\mathcal{P}^{[d]}) \cong \mathcal{P}^{[\deg \mathfrak{m}]} \boxtimes \mathcal{P}^{[d-\deg \mathfrak{m}]}$ from Proposition 24, the tame ramification of the box product at θ descends to the tame ramification of $\mathcal{P}^{[d]}$ at η_0 by applying Lemma 43.

□

Lemma 60. *Let $\mathfrak{n}_1, \mathfrak{n}_2 \subset \mathfrak{m}$ be two coprime sub moduli of $\mathfrak{m}$. Assume $\mathcal{P}^{[\deg \mathfrak{n}_1]}$, $\mathcal{P}^{[\deg \mathfrak{n}_2]}$ are at most tamely ramified at $\eta_{\mathfrak{n}_1}$, $\eta_{\mathfrak{n}_2}$ respectively. Then $\mathcal{P}^{[\deg \mathfrak{n}_1 + \deg \mathfrak{n}_2]}$ is at most tamely tamified at $\eta_{\mathfrak{n}_1+\mathfrak{n}_2}$.*

Proof. It follows from Proposition 49 and Proposition 24 □

4.6 Proof of Theorem 50

Proof of Theorem 50. Let $\mathcal{F}$ be as in Theorem 50, $\mathfrak{m} = \sum_{i=1}^{n} k_P P$ with $\deg P = d_P$ Then by Theorem 3 for every $\mathfrak{n} \subset \mathfrak{m}$ of the form $\mathfrak{n} = k_P P$, $\mathcal{F}^{[\deg \mathfrak{n}]}$ is at most tamely ramified at $\eta_{\mathfrak{n}}$. By Lemma 60, $\mathcal{F}^{[\deg \mathfrak{m}]}$ is then at most tamely ramified at $\eta_{\mathfrak{m}}$. And thus by Corollary 59 $\mathcal{F}^{[d]}$ is tamely ramified at the generic point η_0 of E_0 □

5 Proof of Theorem 2

We work over $S = \operatorname{Spec} k$, for k perfect.

By Proposition 8, its equivalent to prove:

Theorem 61. *Let $G = \Lambda^\times$ be a finite abelian group (Λ as before), and let $\mathcal{P}$ be a G-torsor on U, with ramification bounded by $\mathfrak{m}$. Then, for sufficiently large integer d, there exists a unique (up to isomorphism) G-torsor $\mathcal{Q}_d$ on $\mathrm{Pic}^d_{C,\mathfrak{m}}$, such that the pullback of $\mathcal{Q}_d$ by Φ_d is isomorphic to $\mathcal{P}^{[d]}$.*

Proof. We divide the proof into two cases, when $\mathfrak{m} = 0$ and when $\mathfrak{m} > 0$.

Case 1: $\mathfrak{m} = 0$. By Section 4.3, for d large enough, the Abel-Jacobi morphism $\Phi_d : C^{(d)} \to \mathrm{Pic}^d_C$ is proper surjective and smooth, with geometrically connected fibers, each isomorphic to $\mathbb{P}^{d-g}_{k^{sep}}$, Hence by Corollary 52 it induces an exact sequence of etale fundamental groups:

$$\pi_1^{et}(\mathbb{P}^{d-g}_{k^{sep}}) \to \pi_1^{et}(C^{(d)}) \to \pi_1^{et}(\mathrm{Pic}^d_C) \to 1$$

But $\mathbb{P}^{d-g}_{k^{sep}}$ is simply connected ([Ten15] Example 4.9, [Tót11] Example 1.4.12), hence its etale fundamental group is trivial, and we get an isomorphism of etale fundamental groups:

$$\pi_1^{et}(C^{(d)}) \cong \pi_1^{et}(\mathrm{Pic}^d_C)$$

Implying the theorem in this case.

Case 2: $\mathfrak{m} > 0$. In this case by Theorem 54, for d large enough, the Abel-Jacobi morphism $\Phi_d : U^{(d)} \to \mathrm{Pic}^d_{C,\mathfrak{m}}$ extends to a proper surjective and smooth map, with geometrically connected fibers isomorphic to projective spaces,

$$\tilde{\Phi}_d : \tilde{C}^{(d)}_{\mathfrak{m}} \to \mathrm{Pic}^d_{C,\mathfrak{m}}$$

Hence we get an isomorphism of etale fundamental groups:

$$\pi_1^{et}(\tilde{C}^{(d)}_{\mathfrak{m}}) \cong \pi_1^{et}(\mathrm{Pic}^d_{C,\mathfrak{m}})$$

By Theorem 50, $\mathcal{F}^{[d]}$ is tamely ramified on the boundary divisor $H = \tilde{C}^{(d)}_{\mathfrak{m}} \setminus U^{(d)}$.

Thus, by Lemma 62 below, we have: $\mathcal{F}^{[d]}$ extends to a locally constant sheaf $\tilde{\mathcal{F}}^{[d]}$ on $\tilde{C}^{(d)}_{\mathfrak{m}}$, which by the isomorphism of etale fundamental groups above, corresponds to a unique locally constant sheaf $\mathcal{G}_d$ on $\mathrm{Pic}^d_{C,\mathfrak{m}}$, such that $\tilde{\Phi}_d^* \mathcal{G}_d \cong \tilde{\mathcal{F}}^{[d]}$. Restricting back to $U^{(d)}$, we get $\Phi_d^* \mathcal{G}_d \cong \mathcal{F}^{[d]}$, as required. □

Lemma 62. *If $\mathcal{F}^{[d]}$ is a locally constant sheaf on $U^{(d)}$ which is tamely ramified along the boundary divisor $H = \tilde{C}^{(d)}_{\mathfrak{m}} \setminus U^{(d)}$, then $\mathcal{F}^{[d]}$ extends to a locally constant sheaf $\tilde{\mathcal{F}}^{[d]}$ on $\tilde{C}^{(d)}_{\mathfrak{m}}$.*

Proof. The lemma we are referencing above can be proved in two routes:
Route 1 - Showing $\ker\{\pi_1^{ab}(U^{(d)}) \to \pi_1^{ab}(\mathrm{Pic}^d_{C,\mathfrak{m}})\}$ is pro-p group.
Route 2 - Showing

$$\pi_1^{t,ab}(U^{(d)}) \to \pi_1^{t,ab}(\mathrm{Pic}^d_{C,\mathfrak{m}})$$

is isomorphism to its image. here one needs to be precise.

□

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